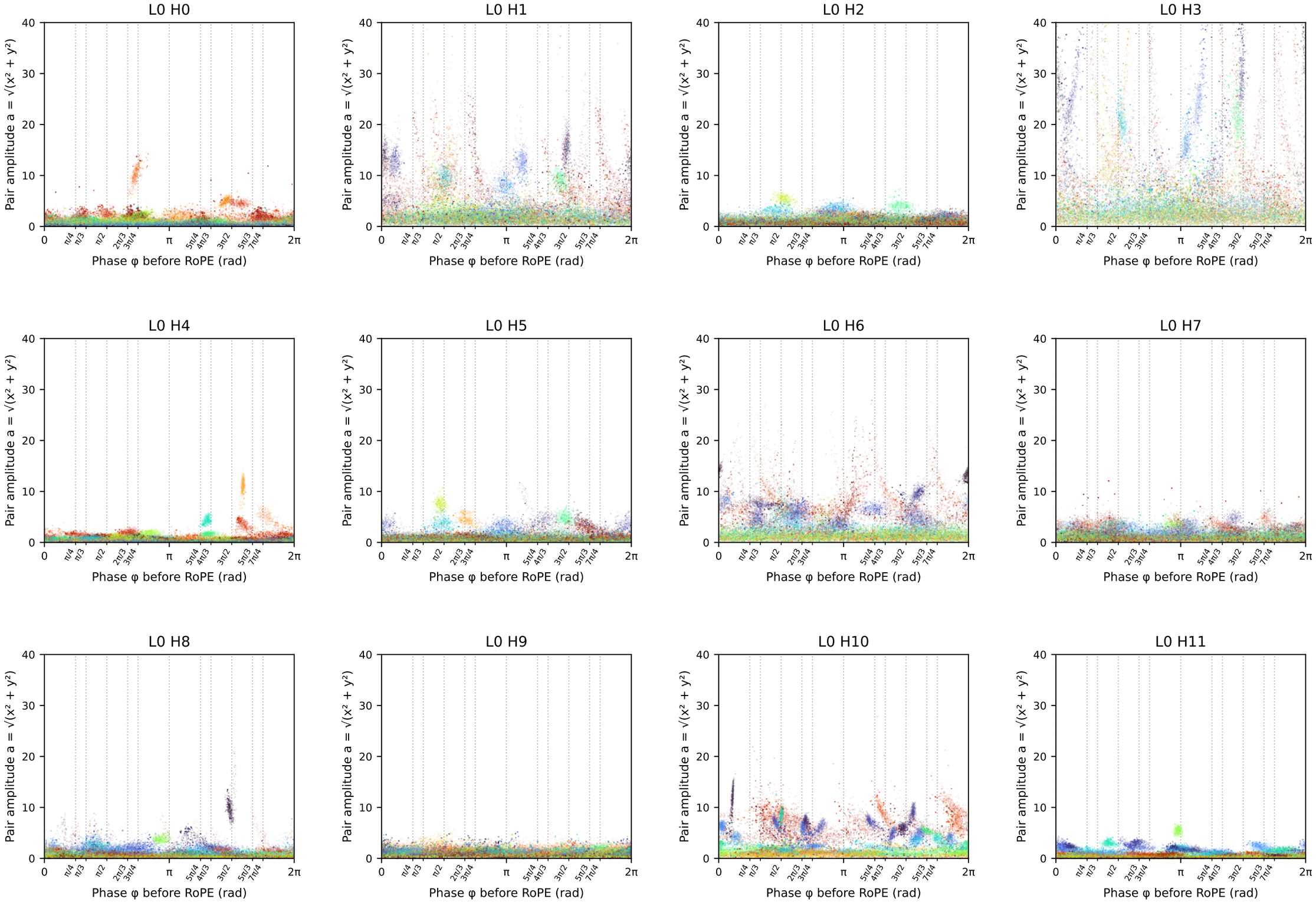
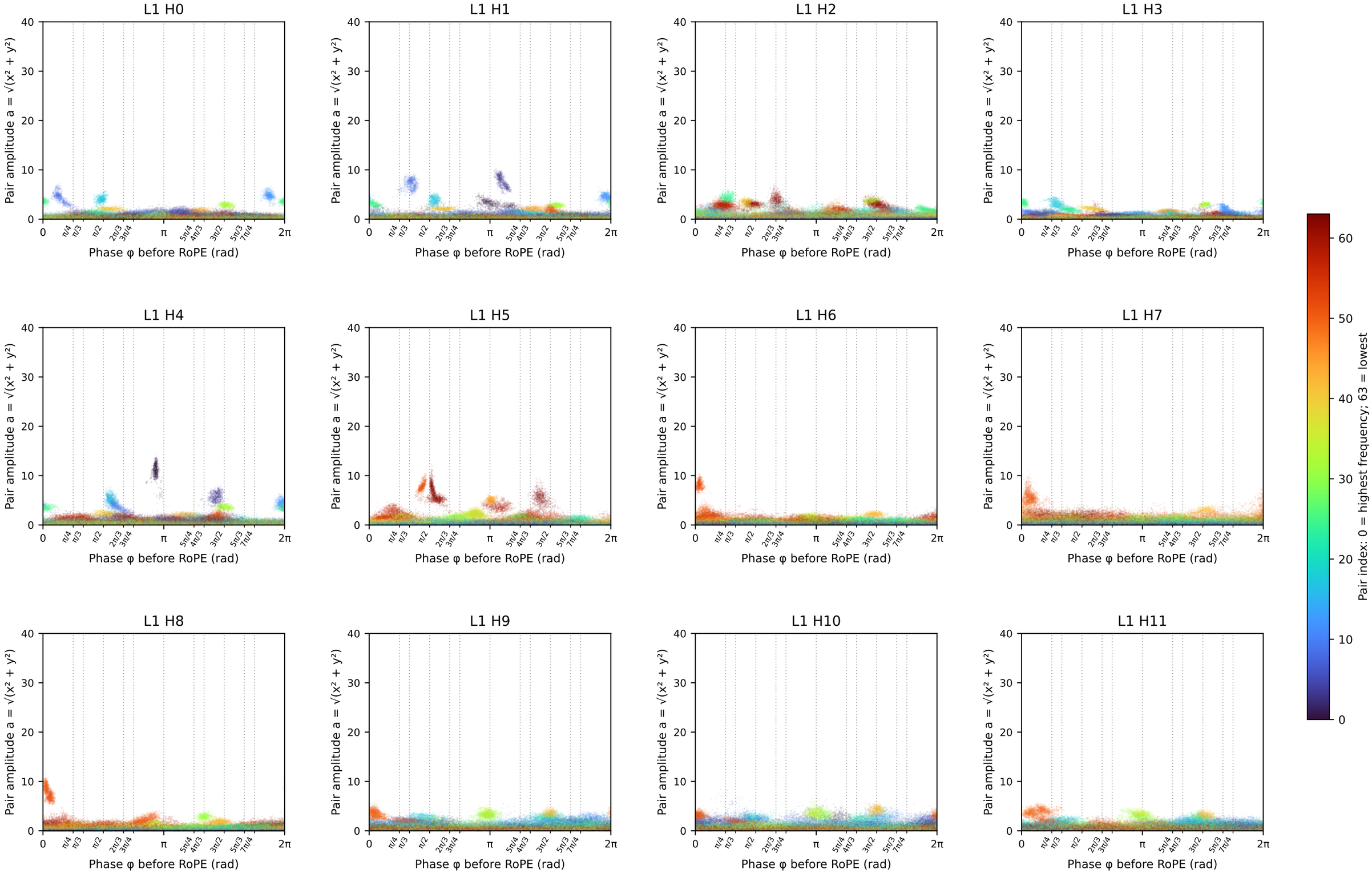
Qwen queries · layer 0 · amplitude axis 0–40



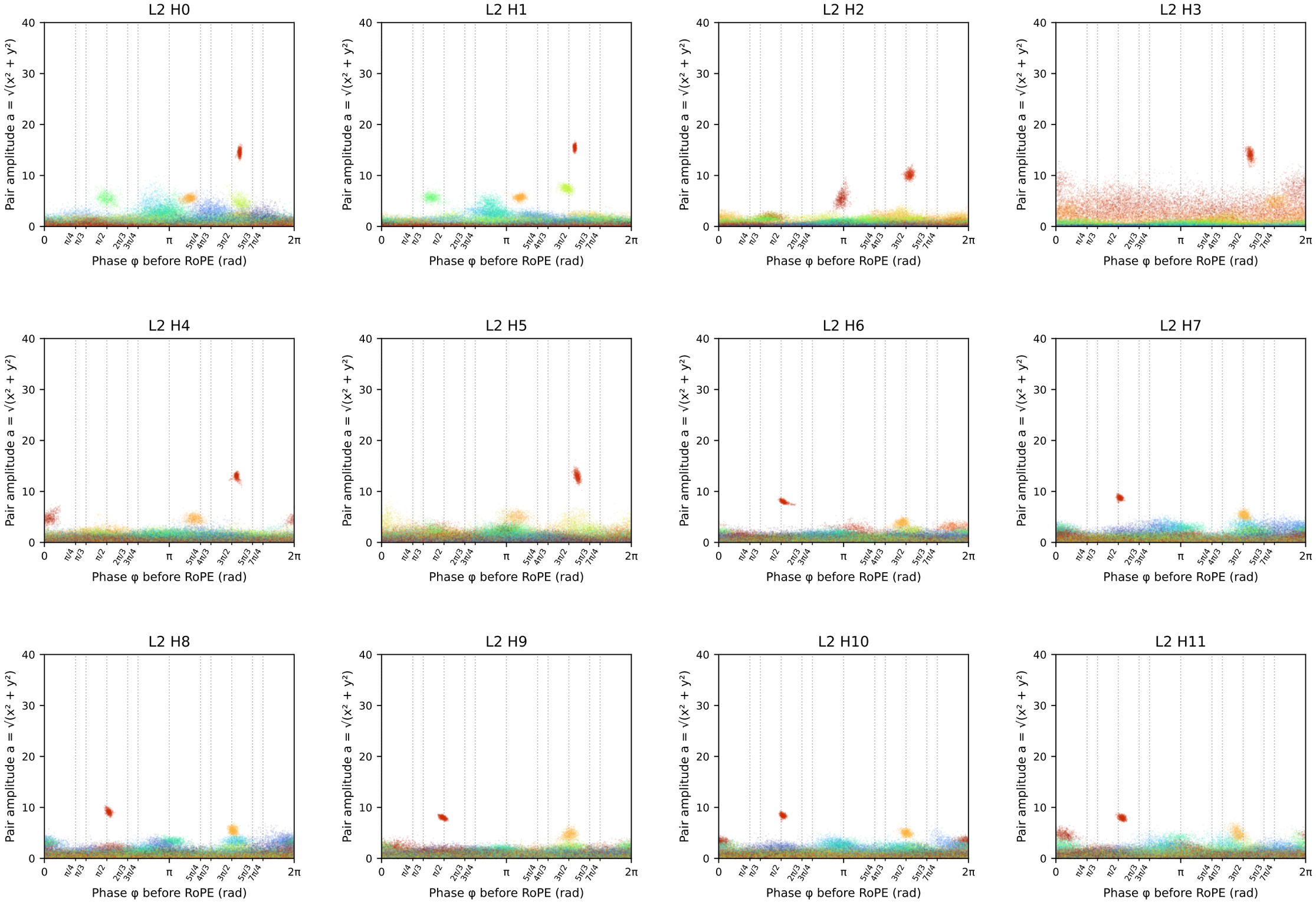
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 1 · amplitude axis 0–40



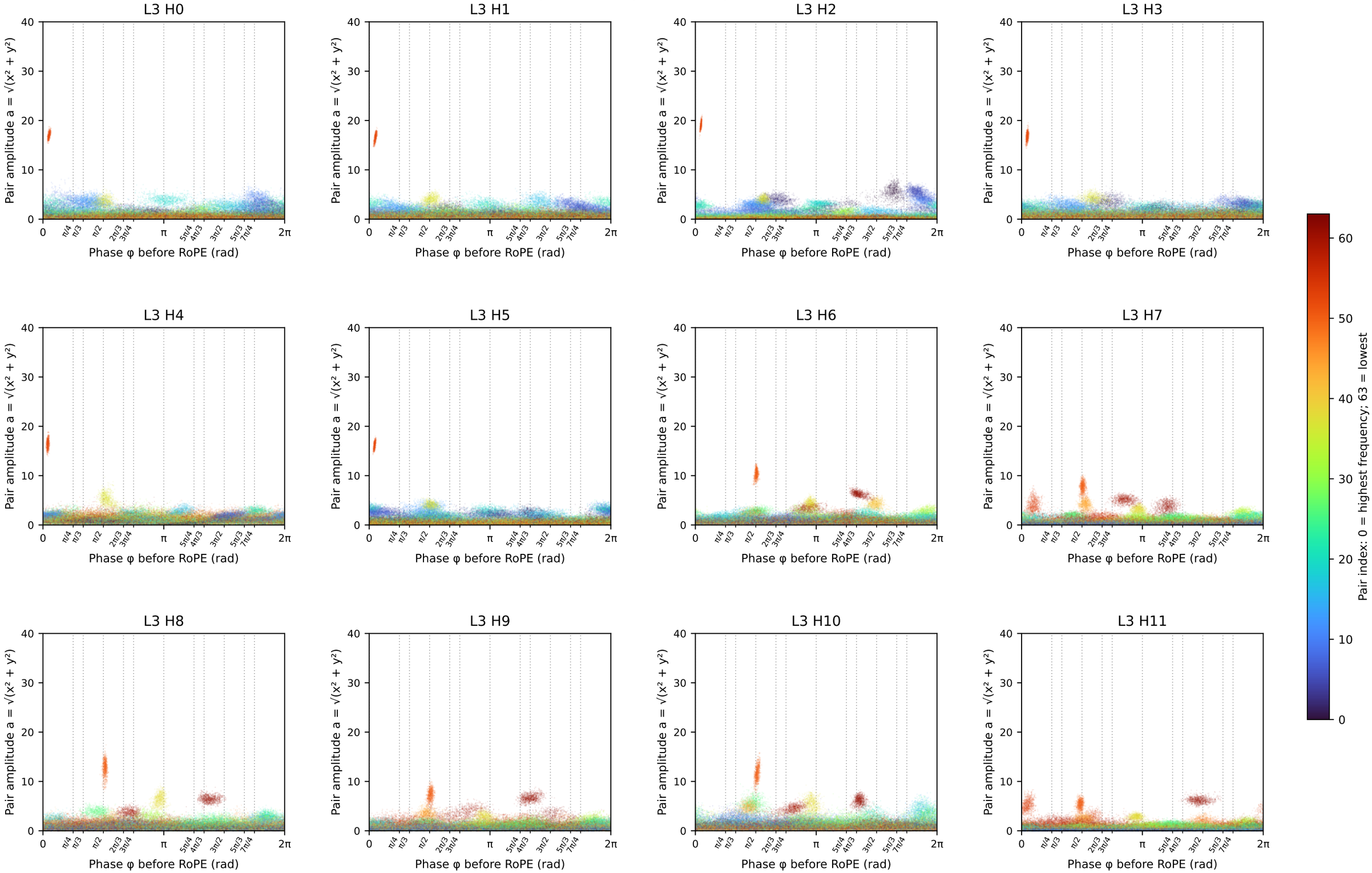
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 2 · amplitude axis 0–40



Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

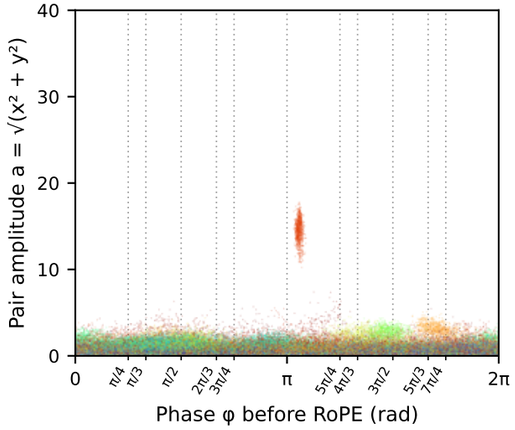
Qwen queries · layer 3 · amplitude axis 0–40



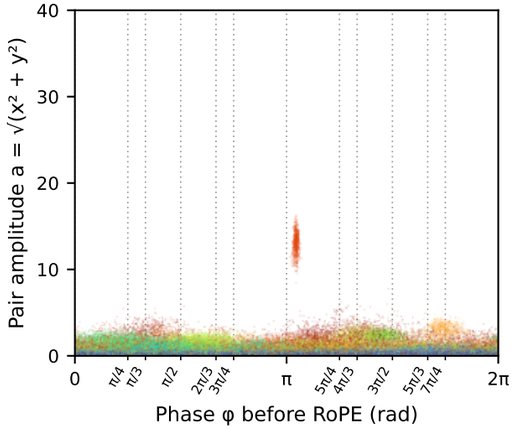
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 4 · amplitude axis 0–40

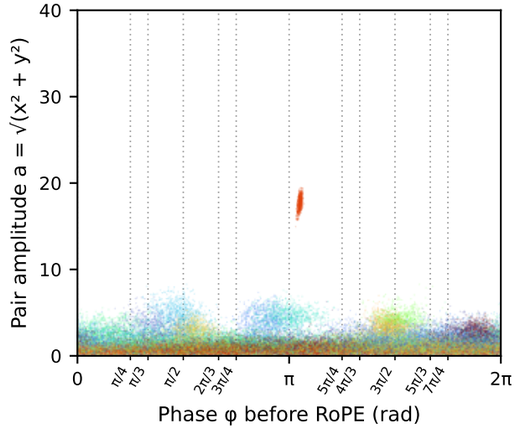
L4 H0



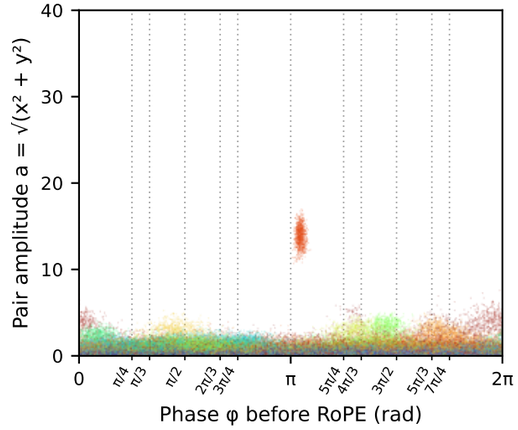
L4 H1



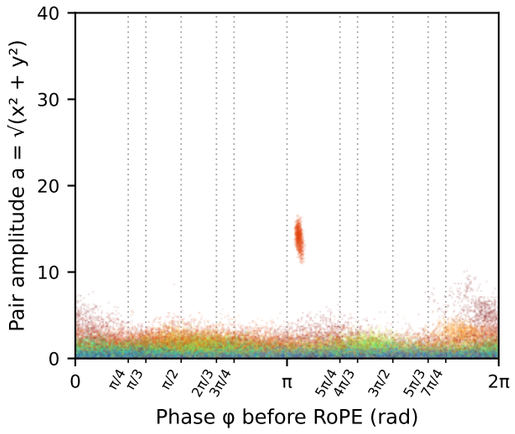
L4 H2



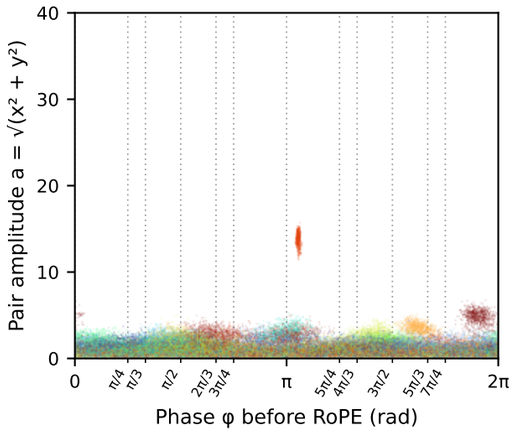
L4 H3



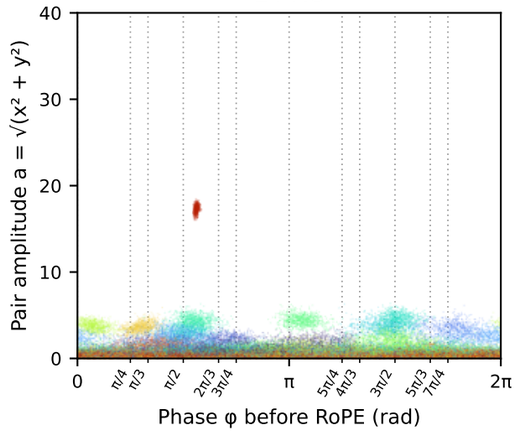
L4 H4



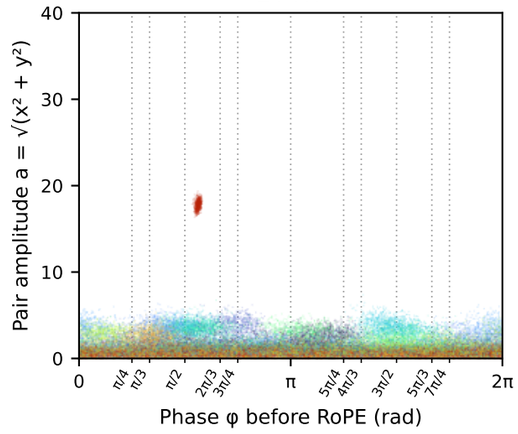
L4 H5



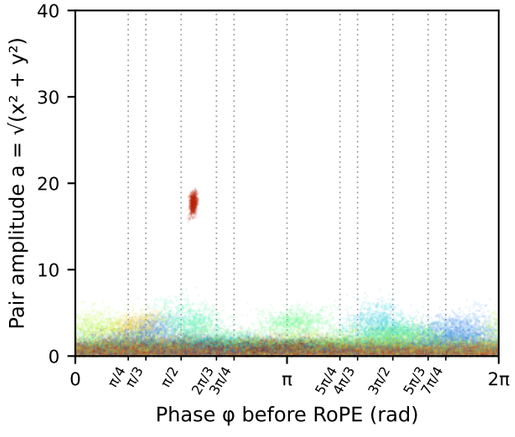
L4 H6



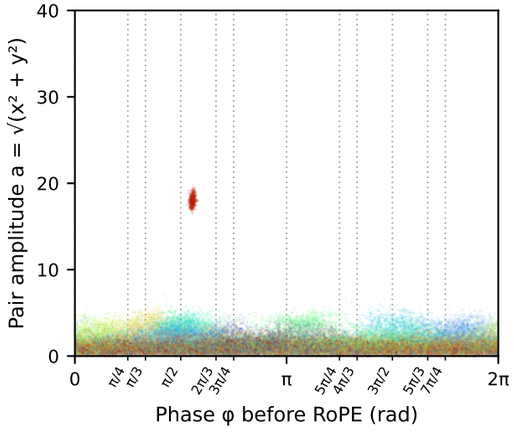
L4 H7



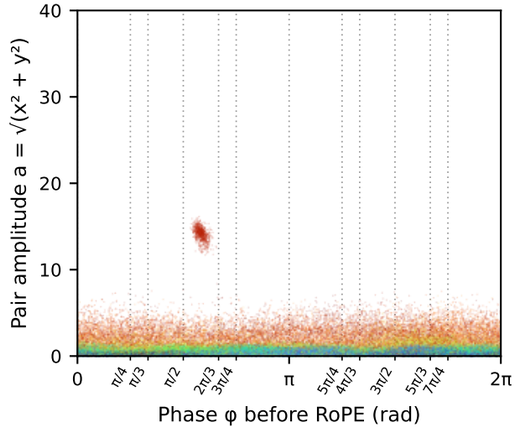
L4 H8



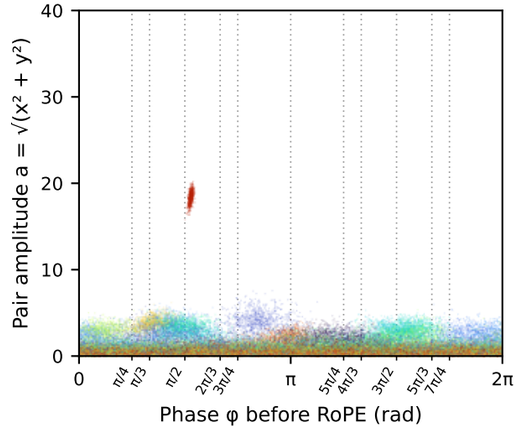
L4 H9

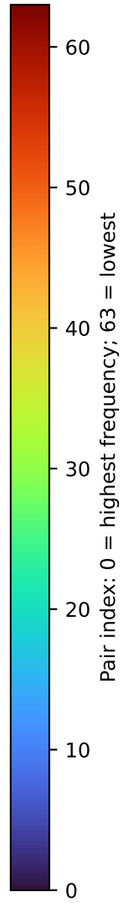


L4 H10



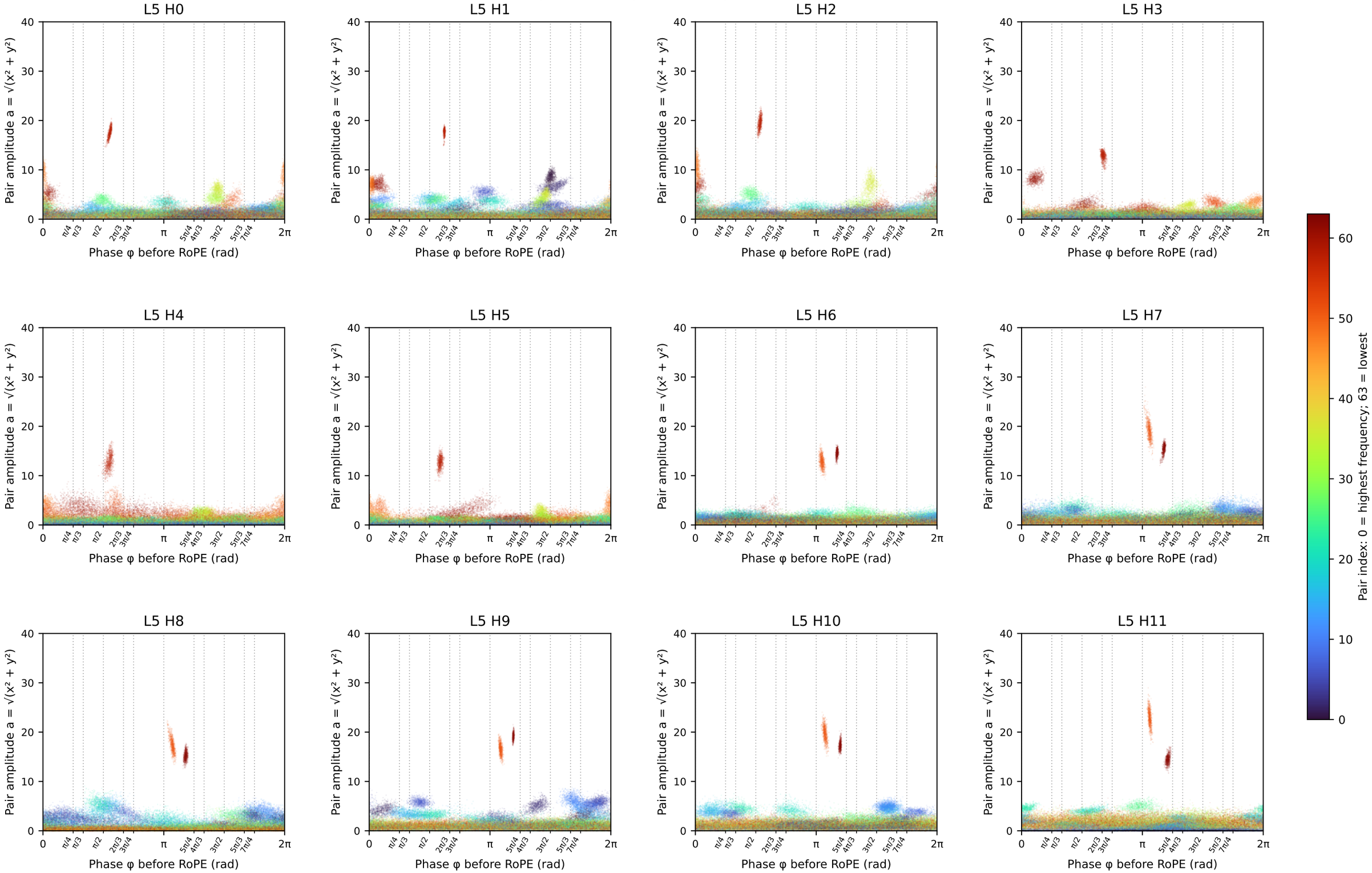
L4 H11





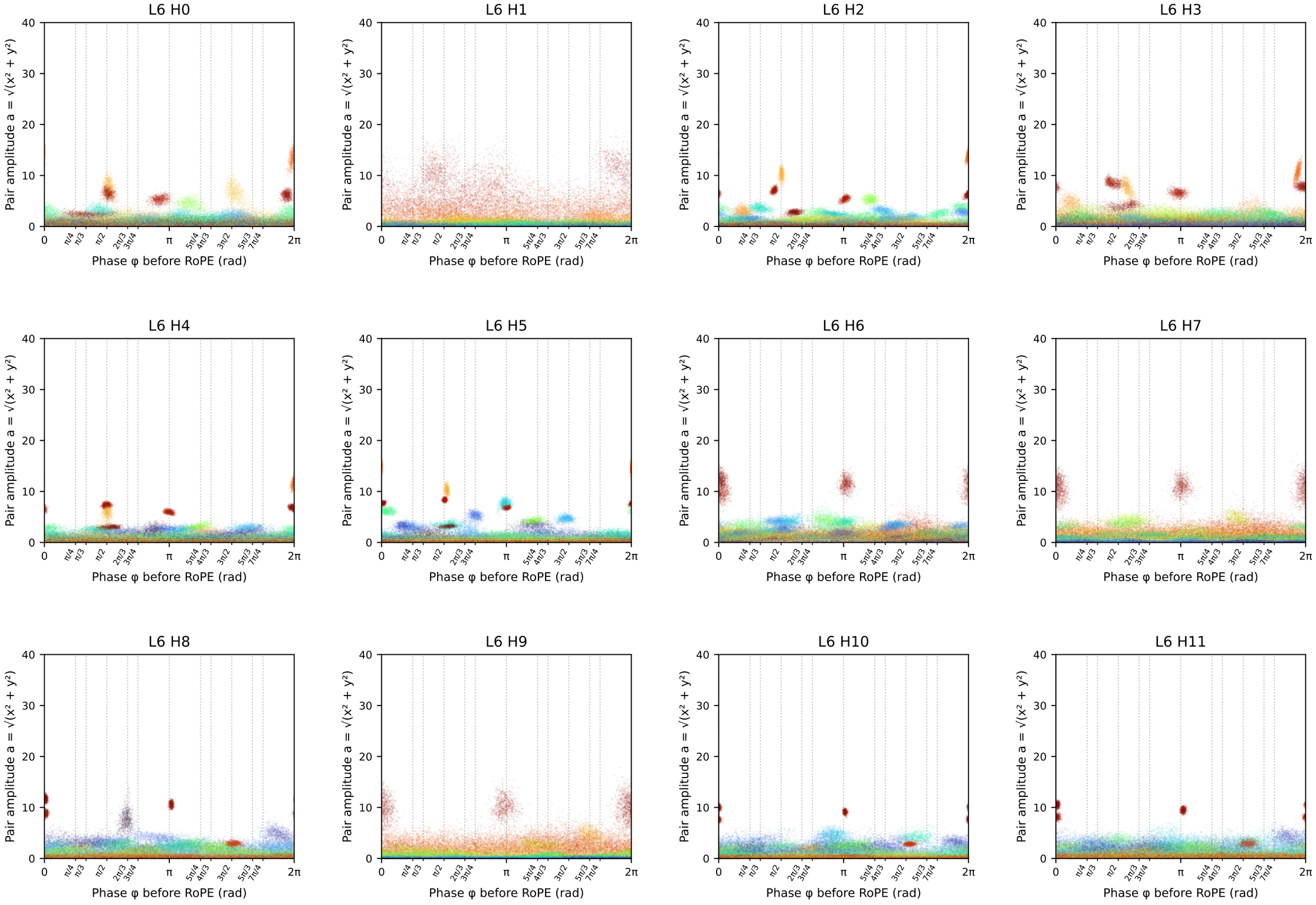
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 5 · amplitude axis 0–40



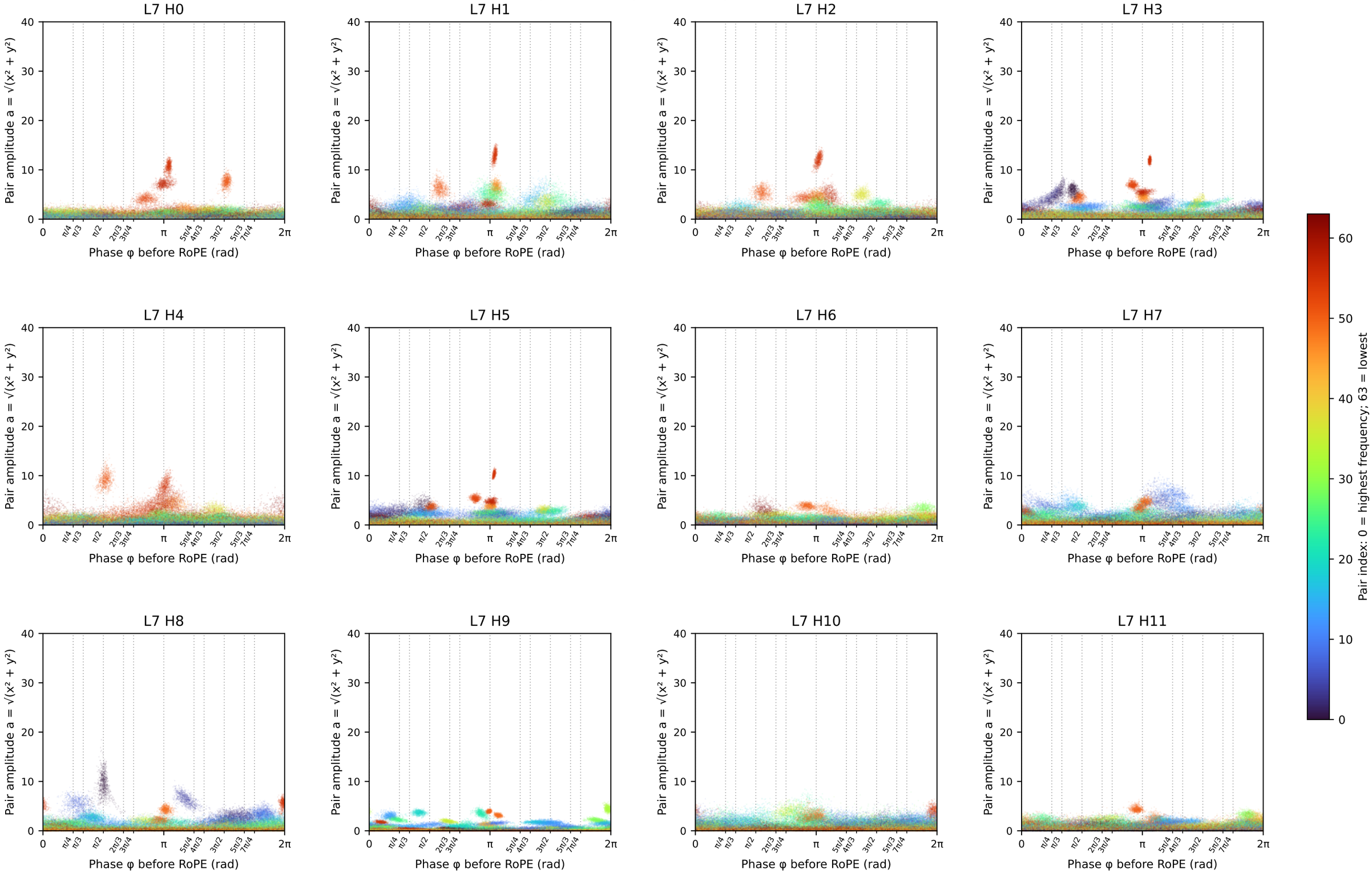
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 6 · amplitude axis 0–40



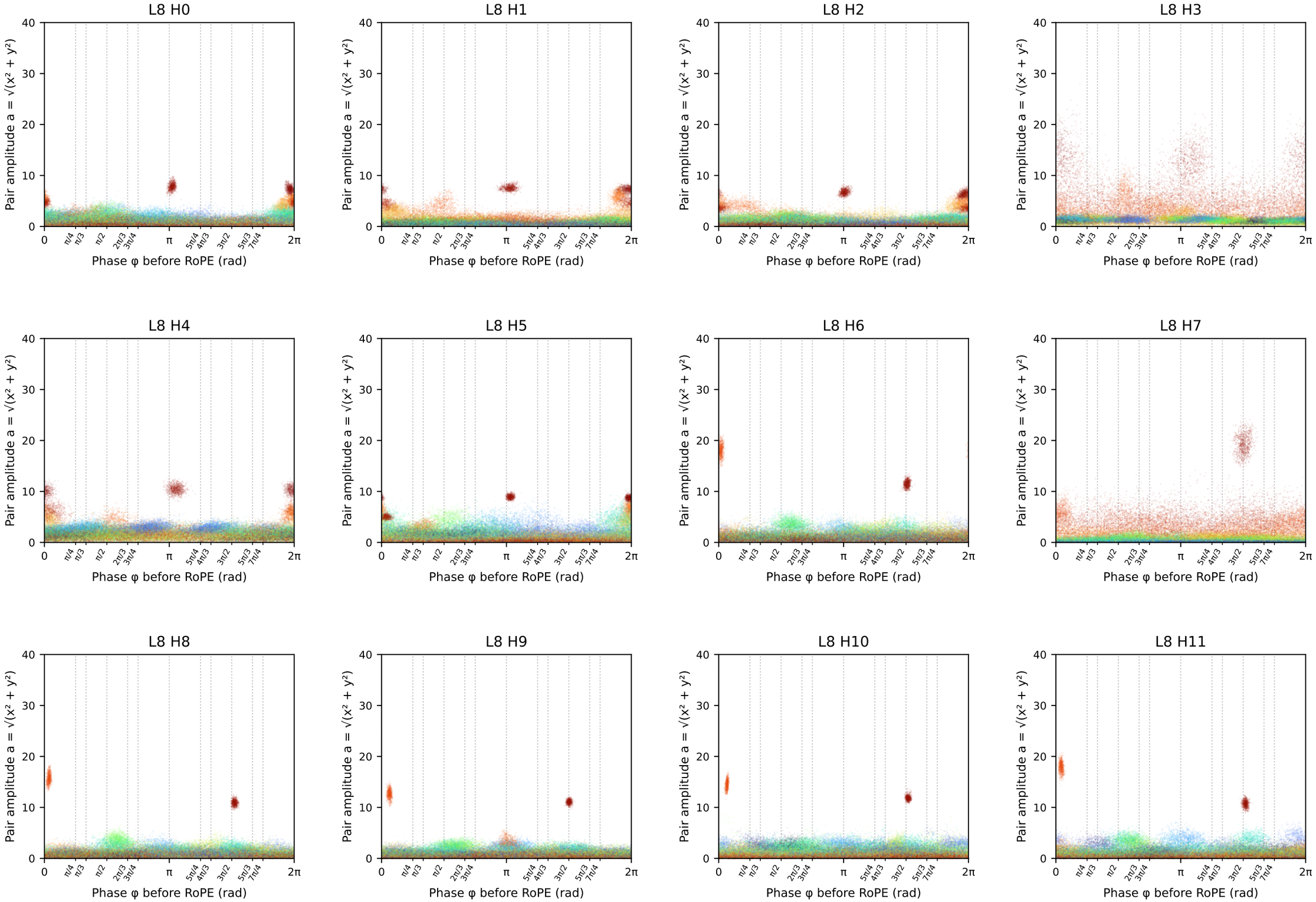
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 7 · amplitude axis 0–40



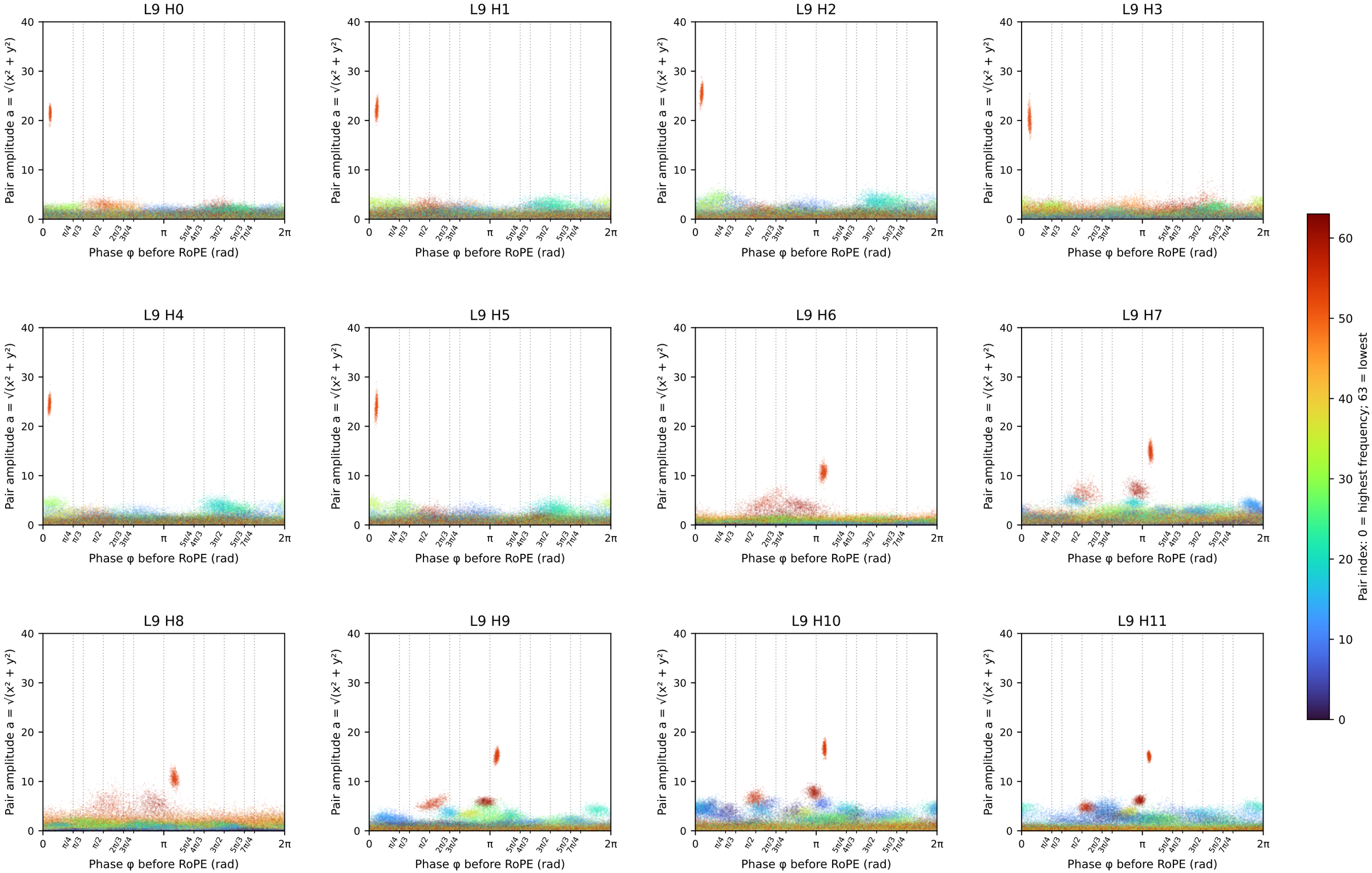
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 8 · amplitude axis 0–40



Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 9 · amplitude axis 0–40



Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 10 · amplitude axis 0–40

L10 H0 | L10 H1 | L10 H2 | L10 H3

L10 H4 | L10 H5 | L10 H6 | L10 H7

L10 H8 | L10 H9 | L10 H10 | L10 H11

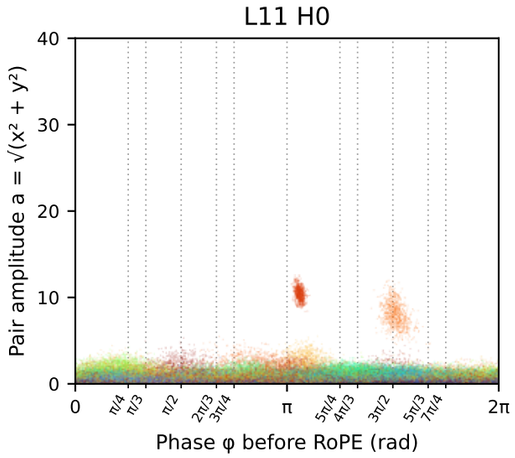
Pair amplitude a = √(x² + y²)

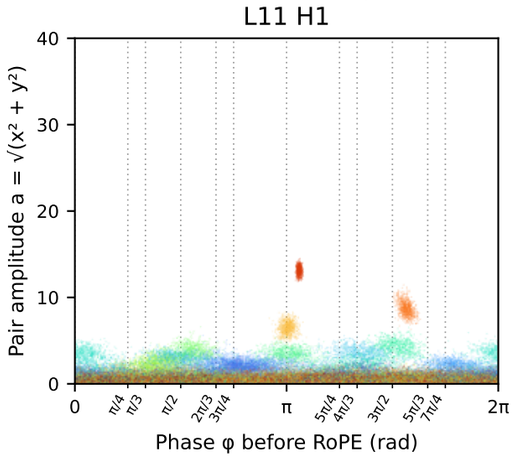
Phase φ before RoPE (rad)

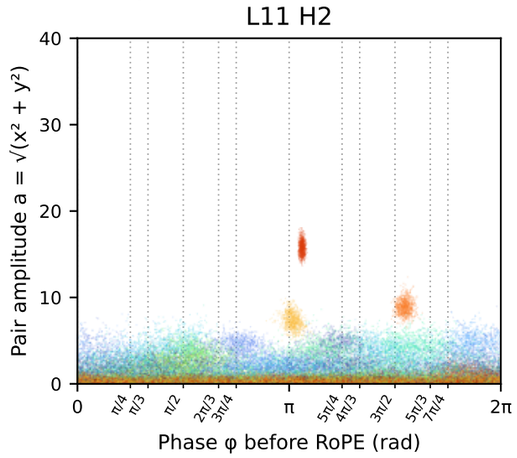
Pair index: 0 = highest frequency; 63 = lowest

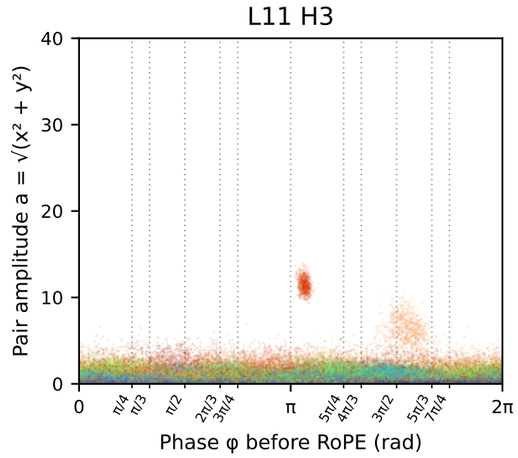
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

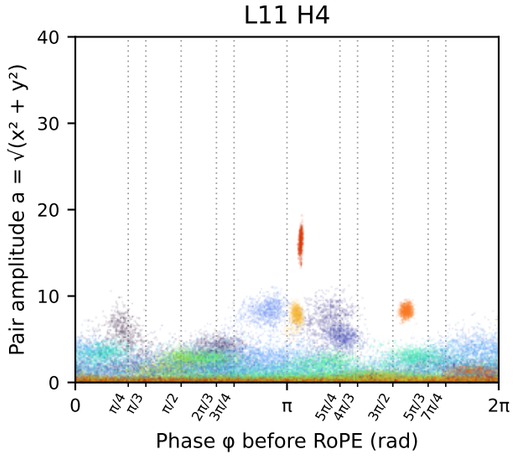
Qwen queries · layer 11 · amplitude axis 0–40

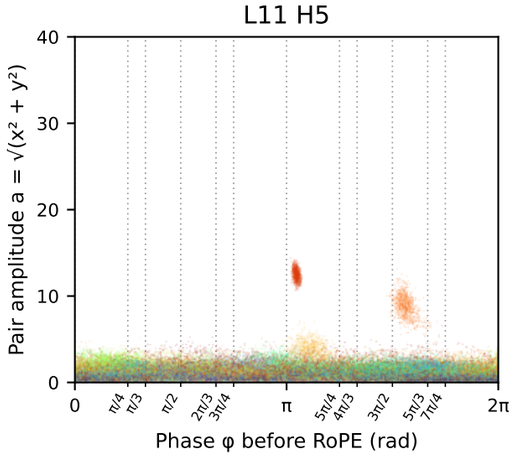


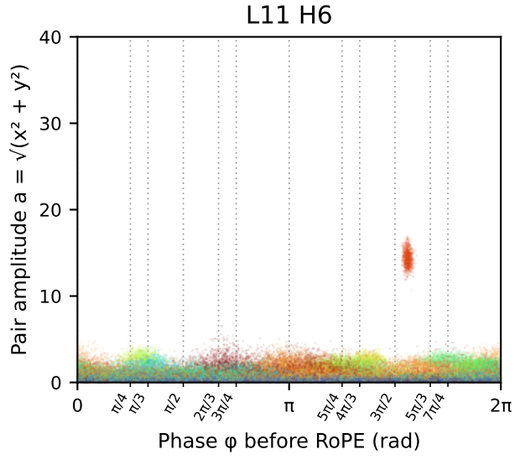


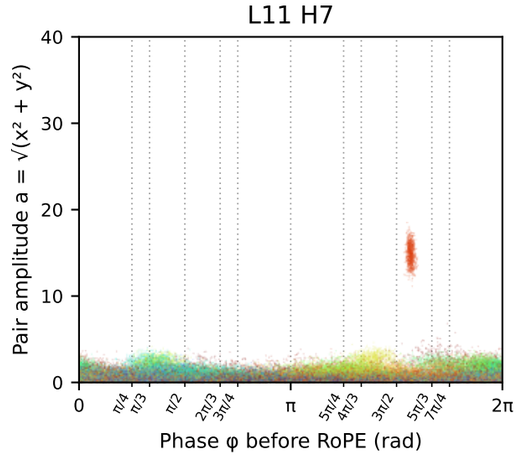


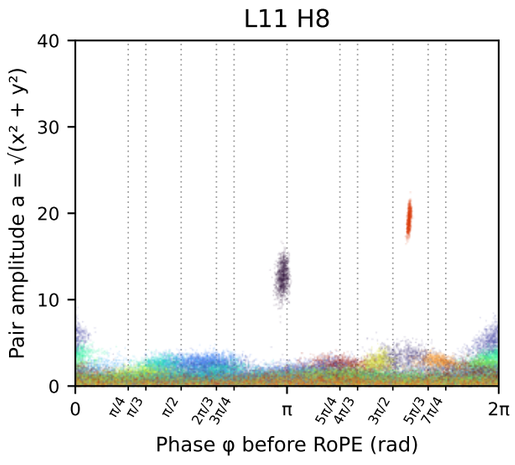


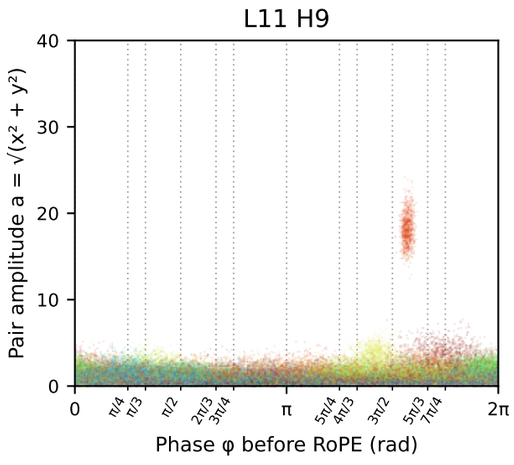


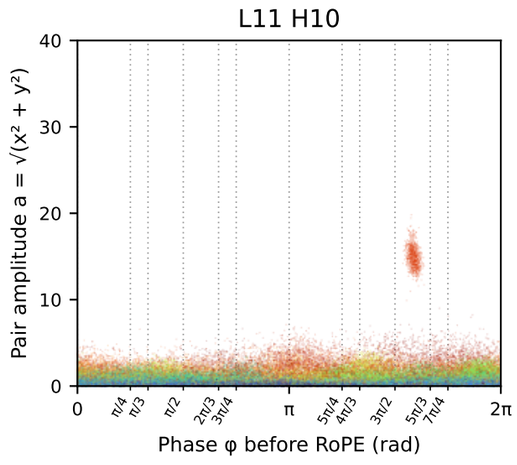


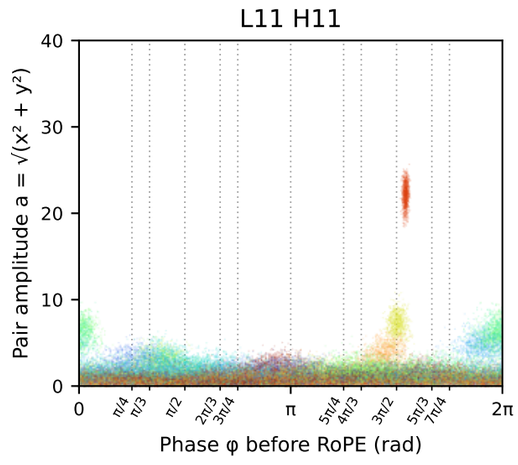


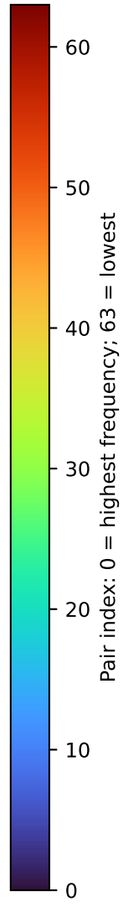








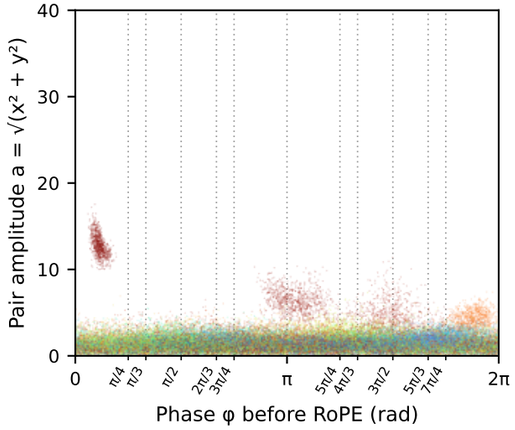




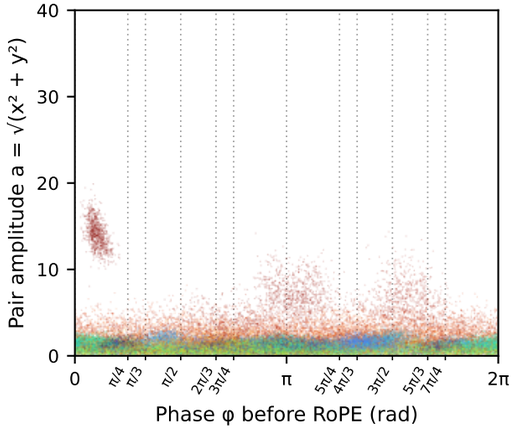
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 12 · amplitude axis 0–40

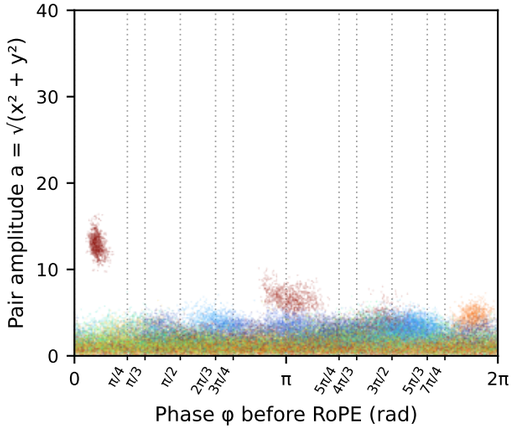
L12 H0



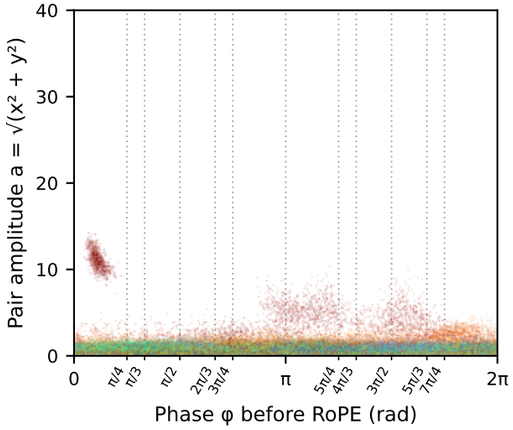
L12 H1



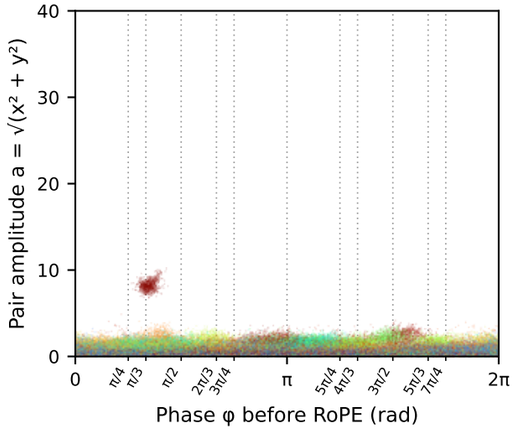
L12 H2



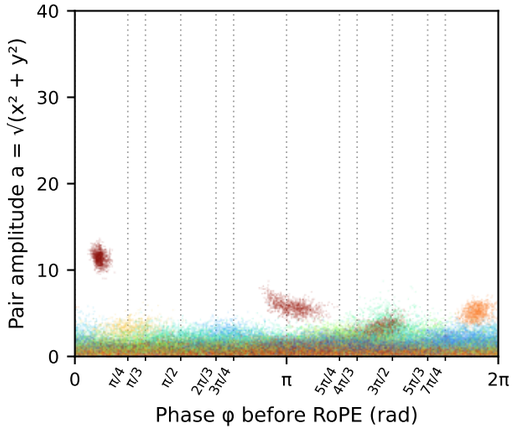
L12 H3



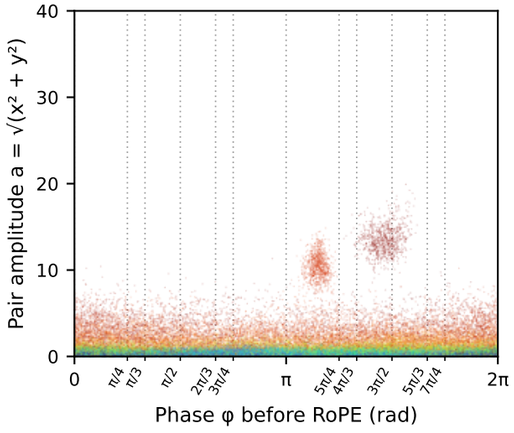
L12 H4



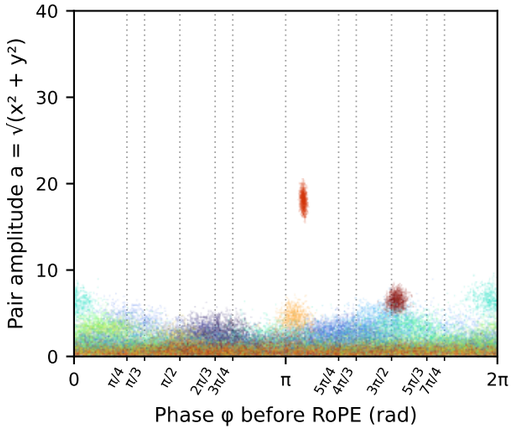
L12 H5



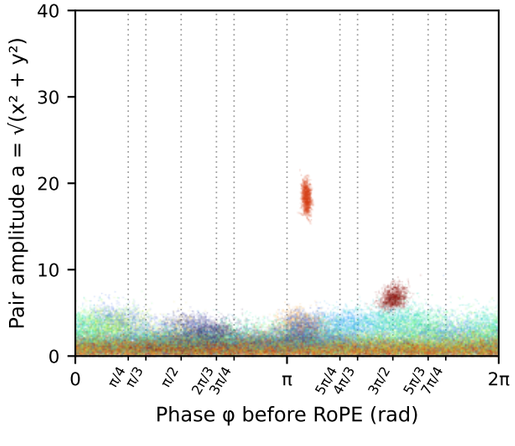
L12 H6



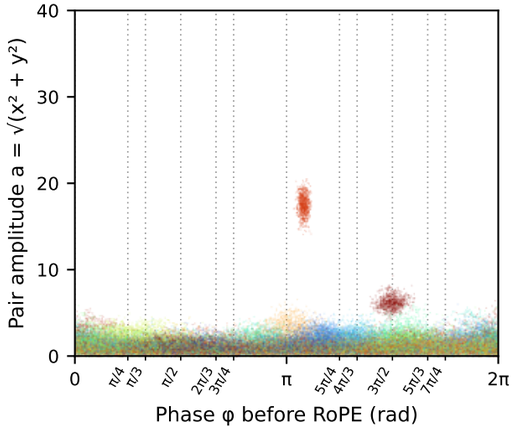
L12 H7



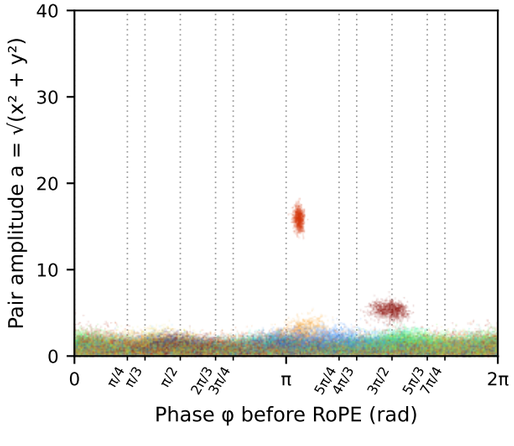
L12 H8



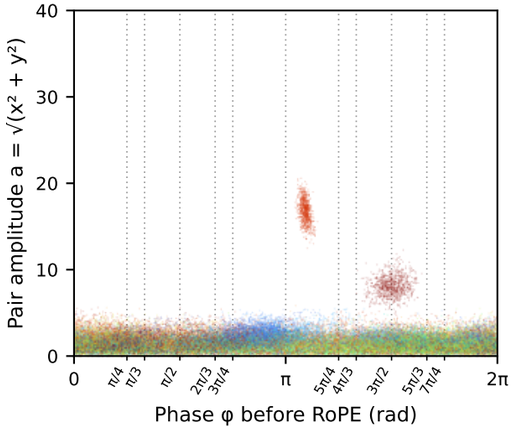
L12 H9

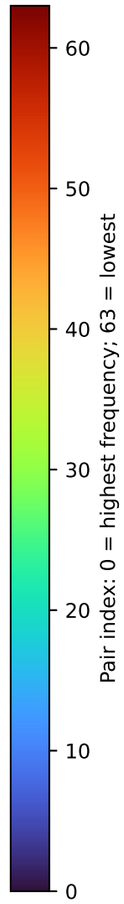


L12 H10



L12 H11

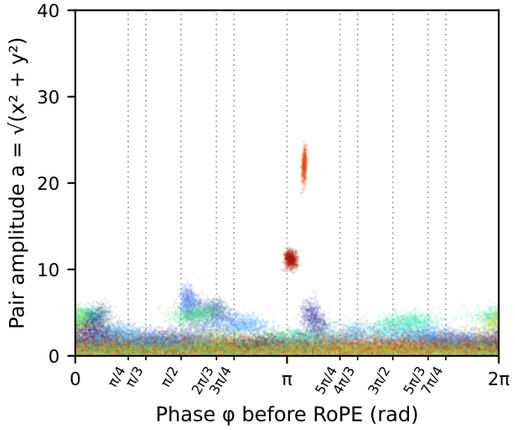




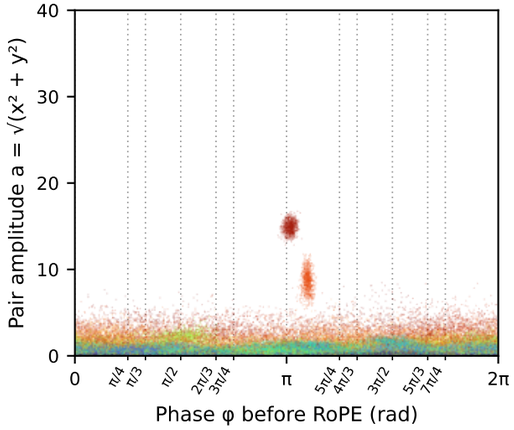
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 13 · amplitude axis 0–40

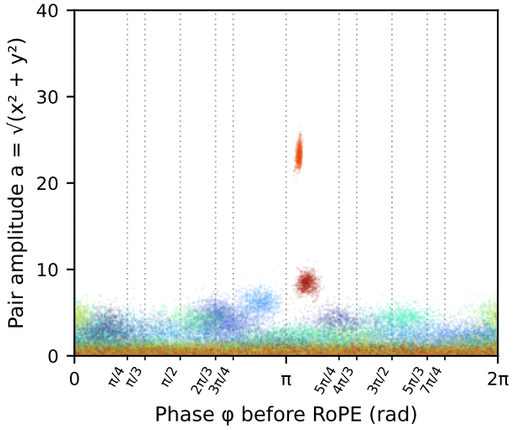
L13 H0



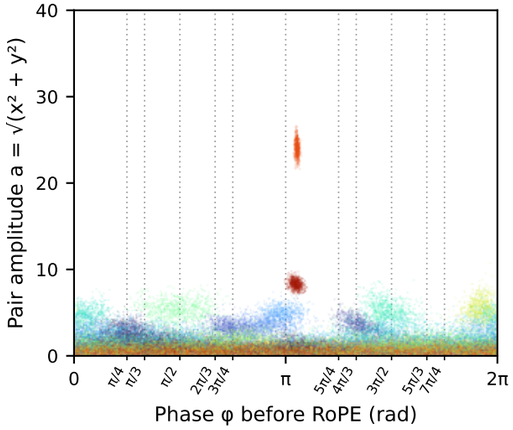
L13 H1



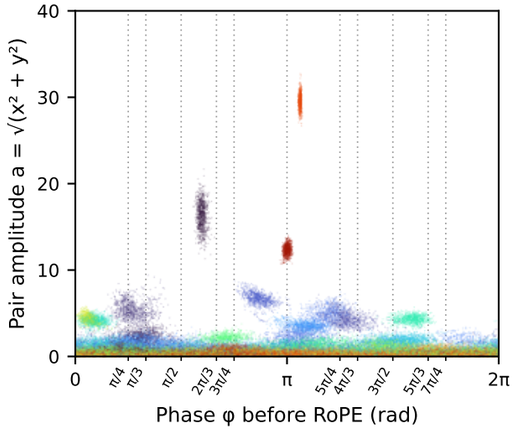
L13 H2



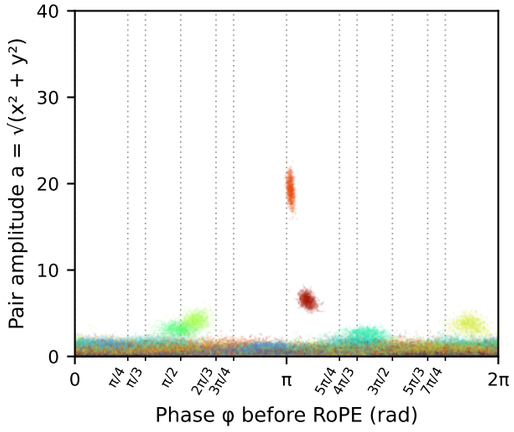
L13 H3



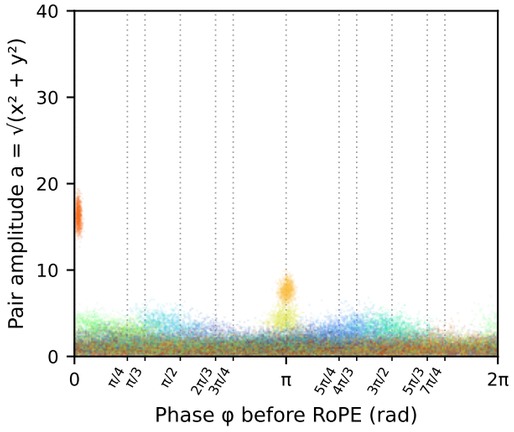
L13 H4



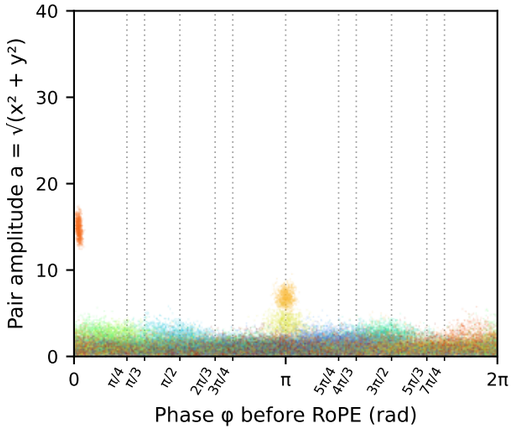
L13 H5



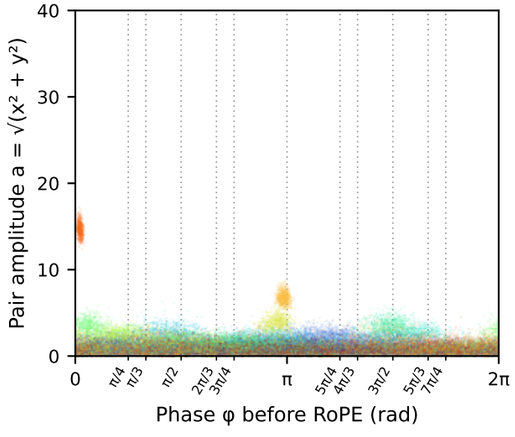
L13 H6



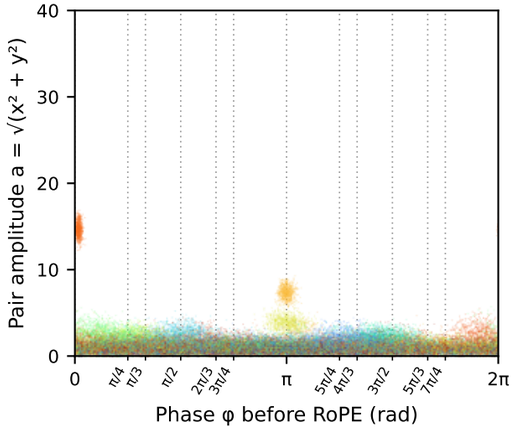
L13 H7



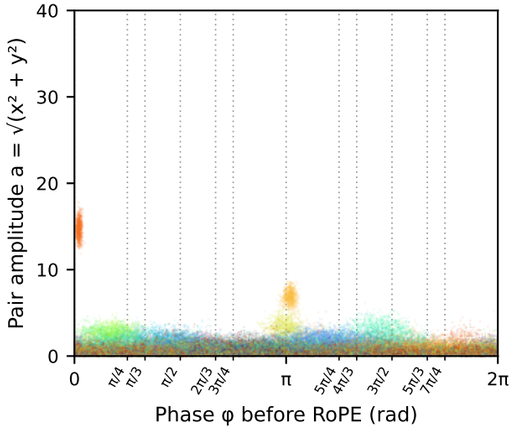
L13 H8



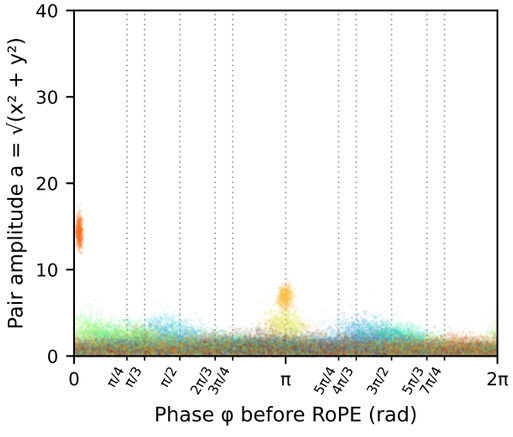
L13 H9

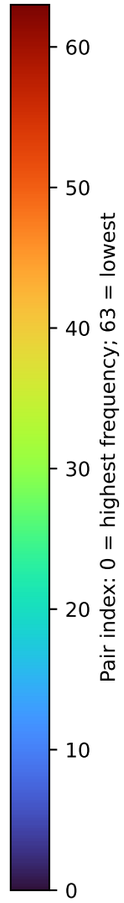


L13 H10



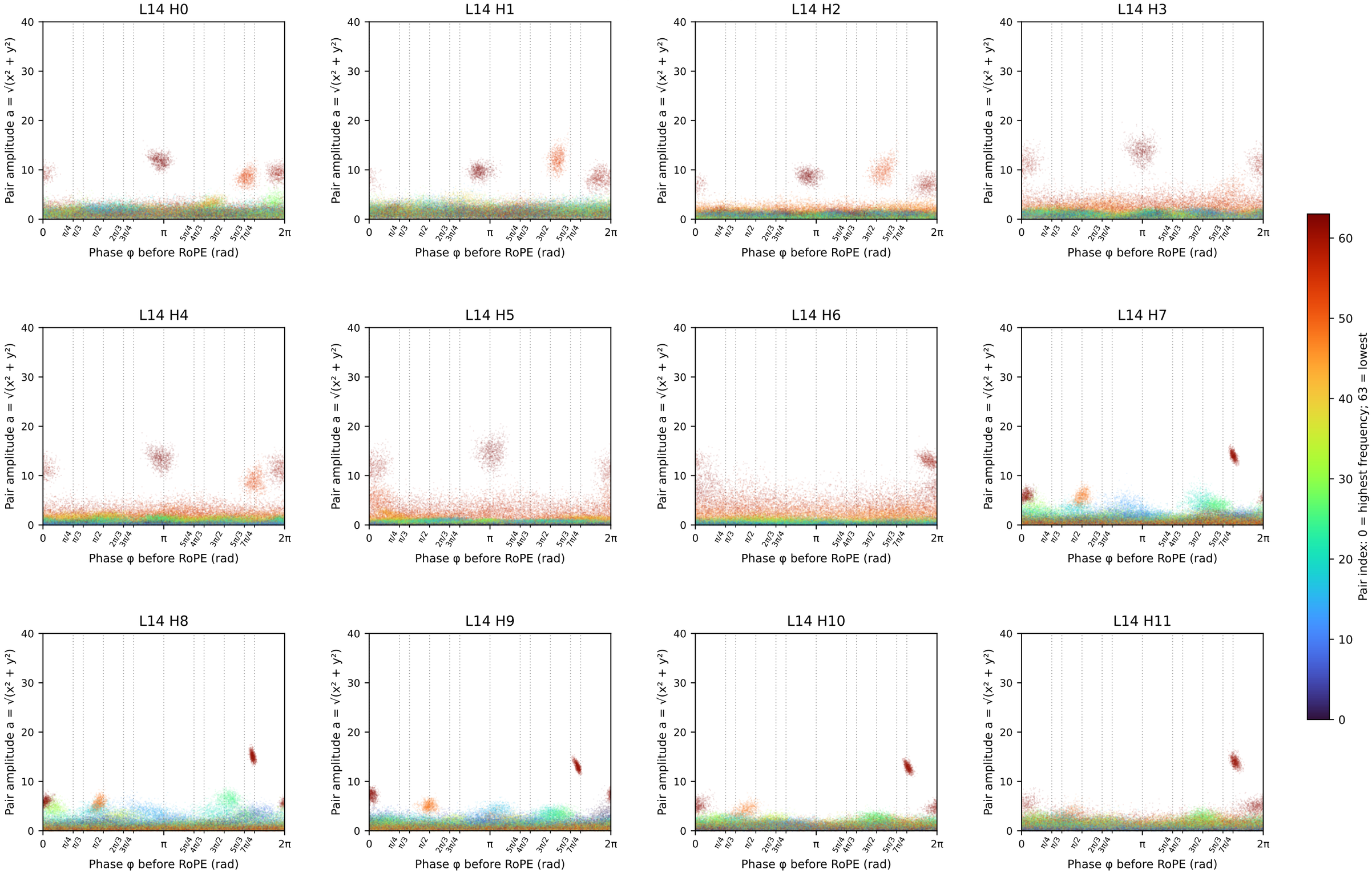
L13 H11





Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

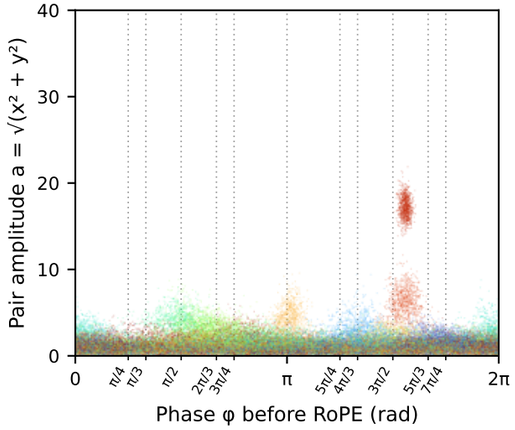
Qwen queries · layer 14 · amplitude axis 0–40



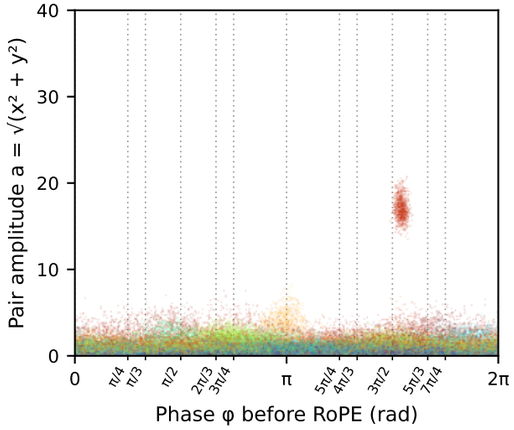
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 15 · amplitude axis 0–40

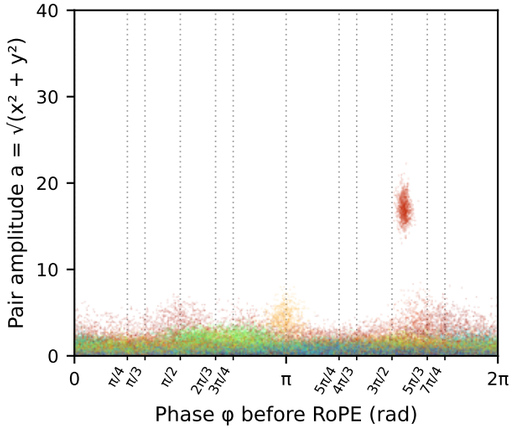
L15 H0



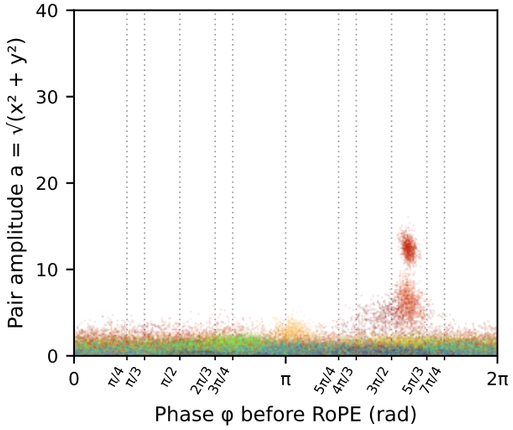
L15 H1



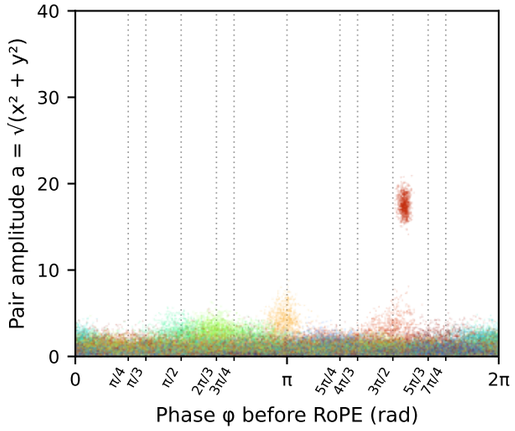
L15 H2



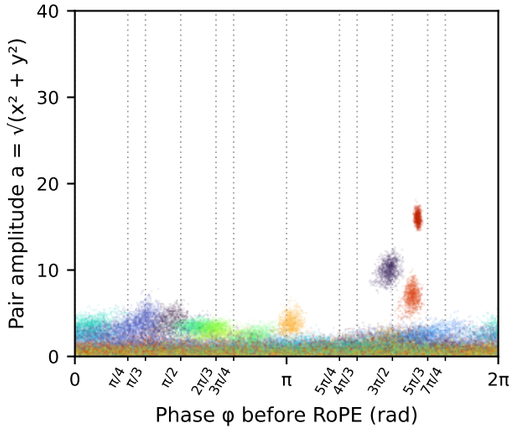
L15 H3



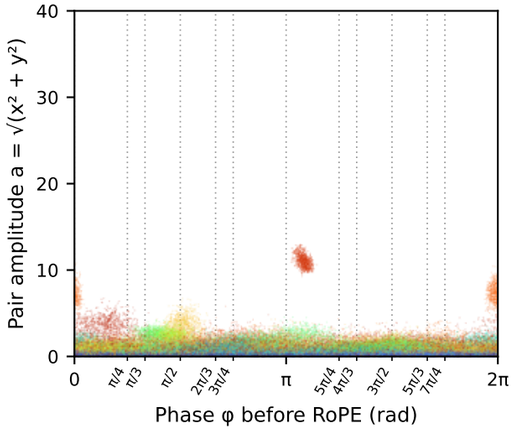
L15 H4



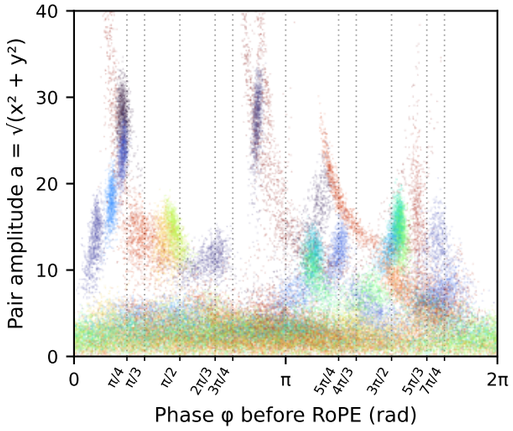
L15 H5



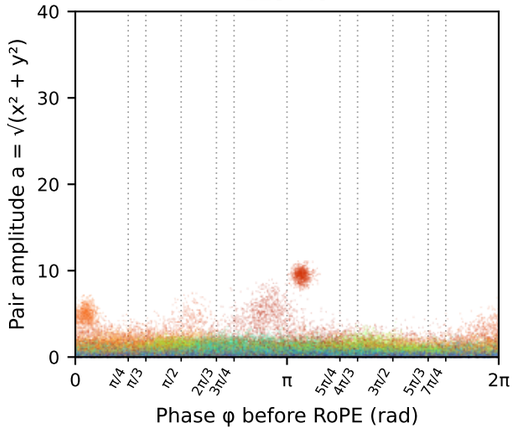
L15 H6



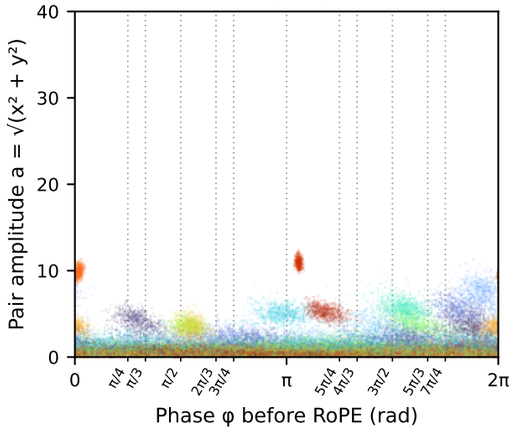
L15 H7



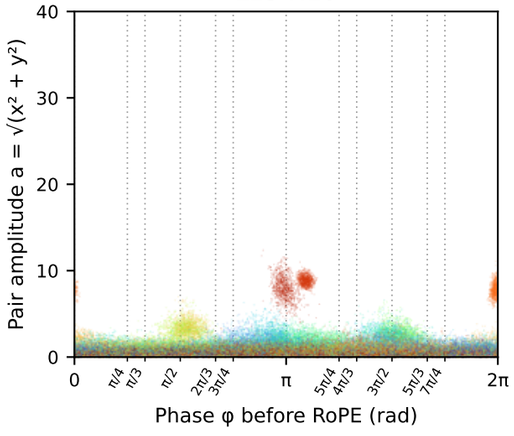
L15 H8



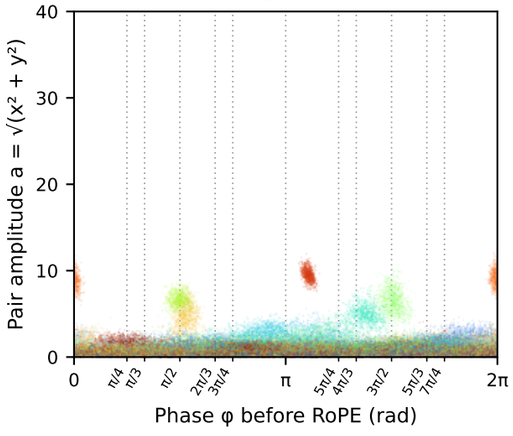
L15 H9

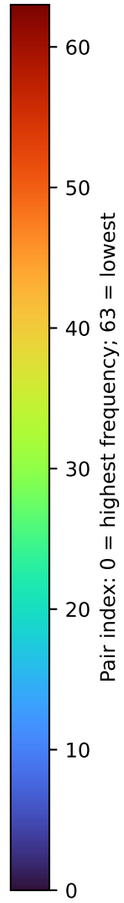


L15 H10



L15 H11

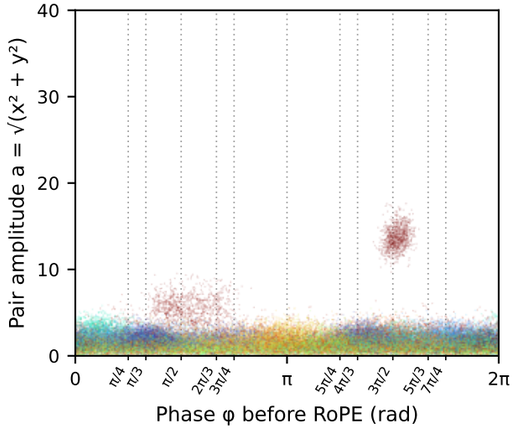




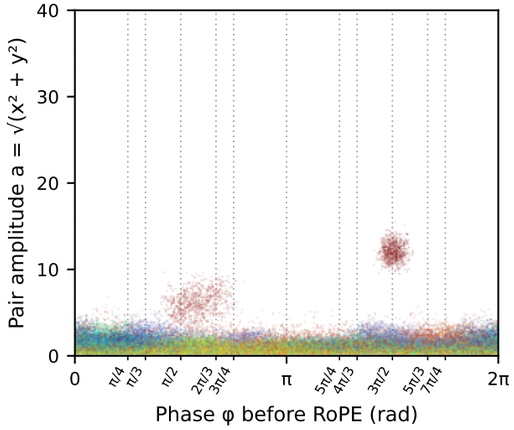
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 16 · amplitude axis 0–40

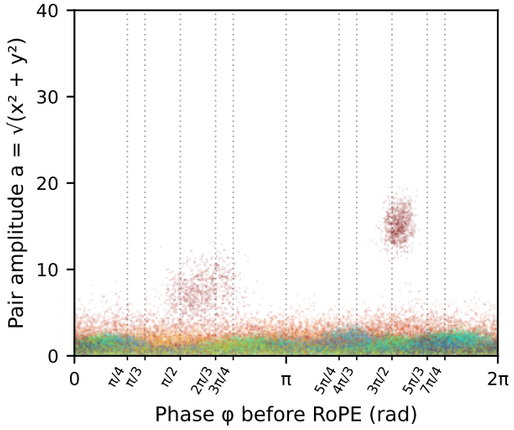
L16 H0



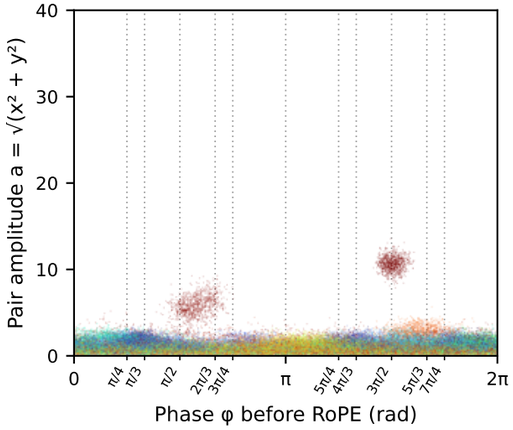
L16 H1



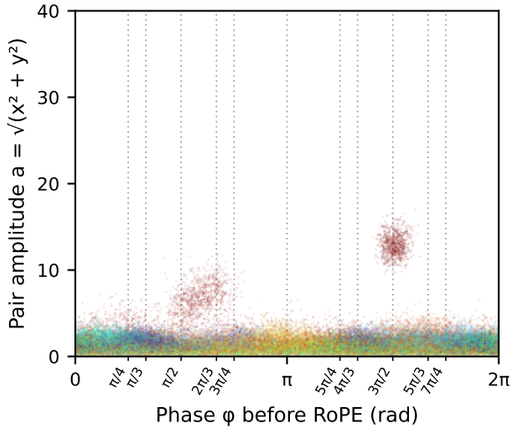
L16 H2



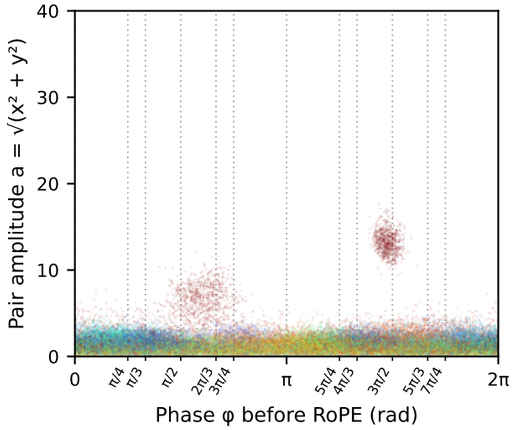
L16 H3



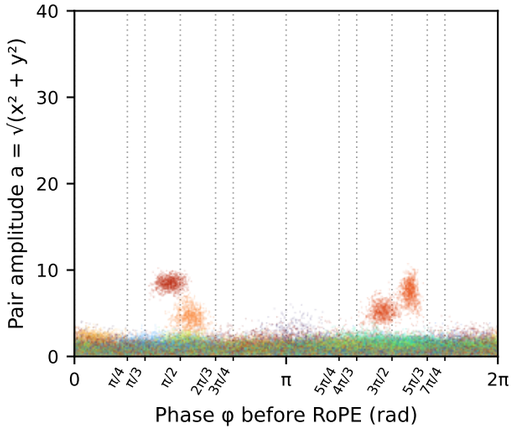
L16 H4



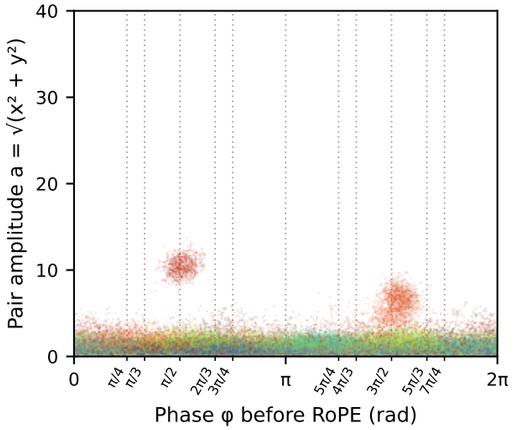
L16 H5



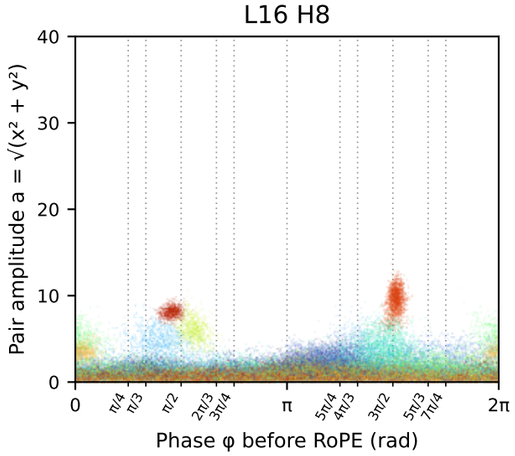
L16 H6



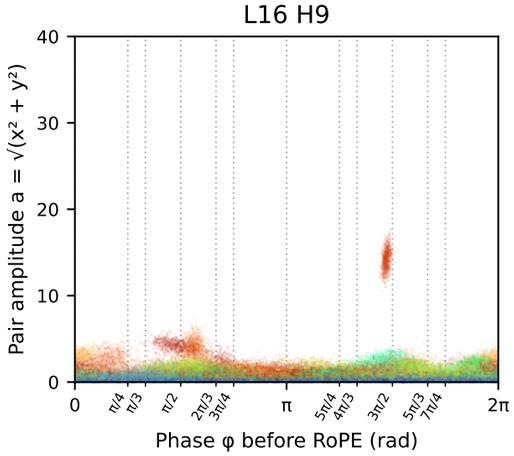
L16 H7



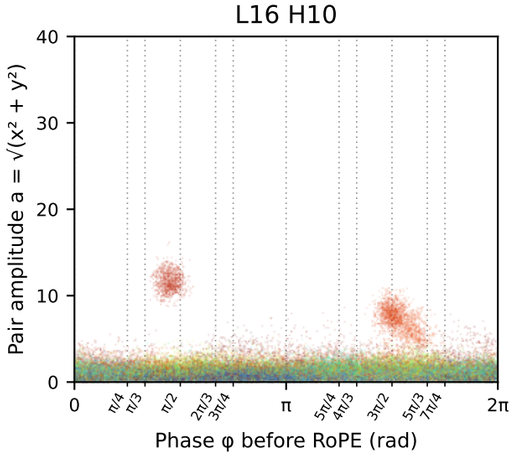
L16 H8



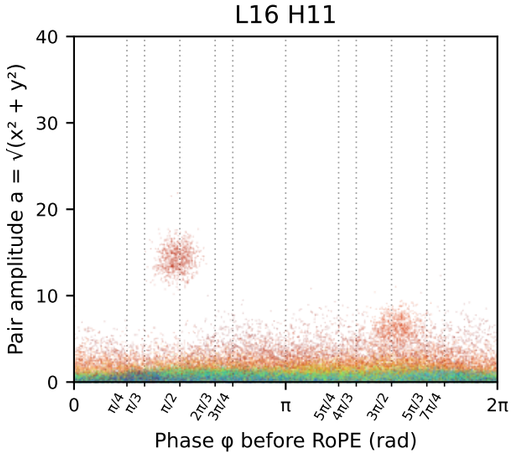
L16 H9

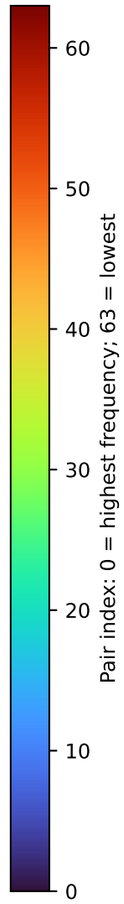


L16 H10



L16 H11

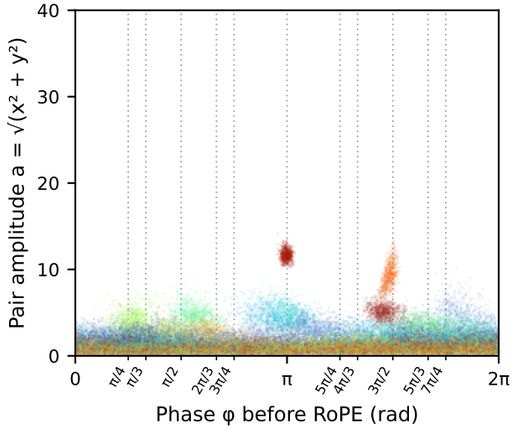




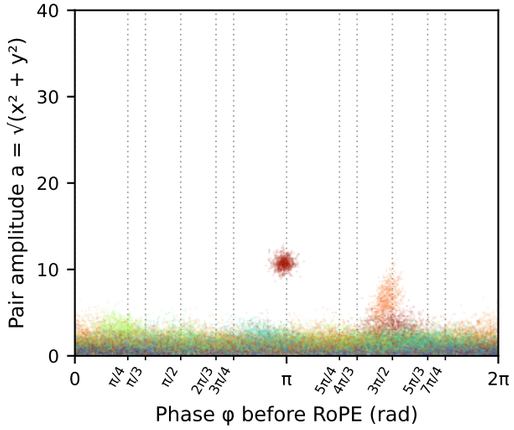
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 17 · amplitude axis 0–40

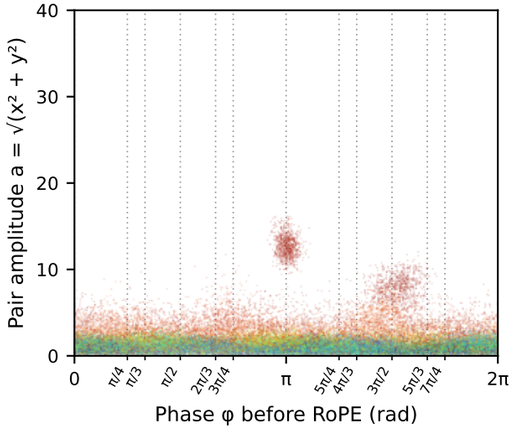
L17 H0



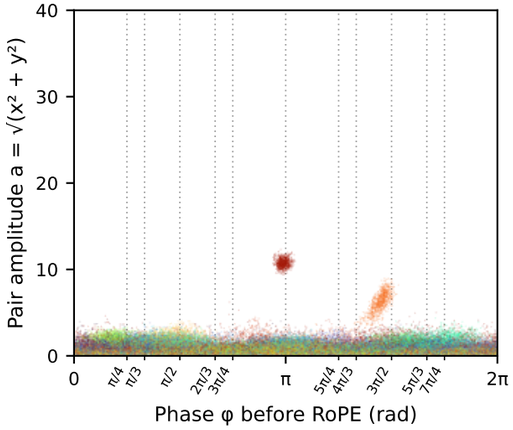
L17 H1



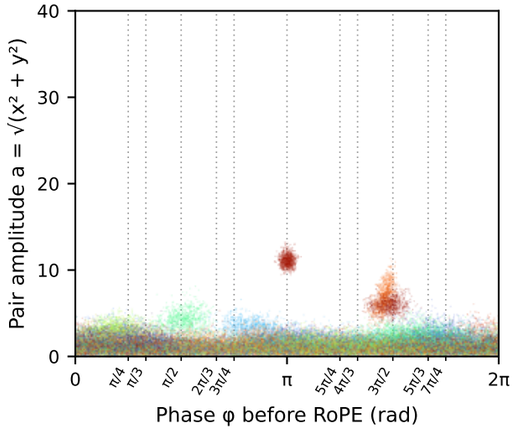
L17 H2



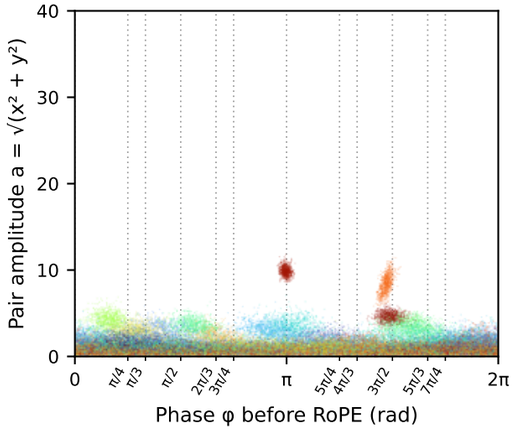
L17 H3



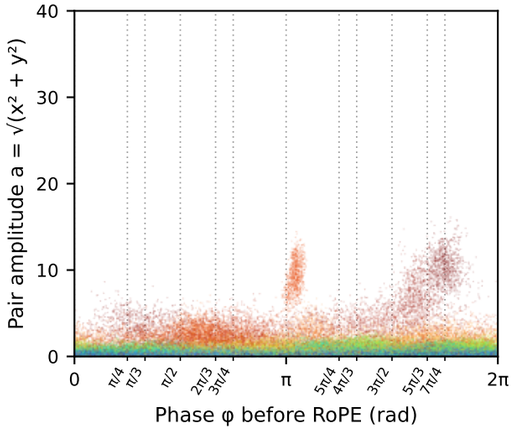
L17 H4



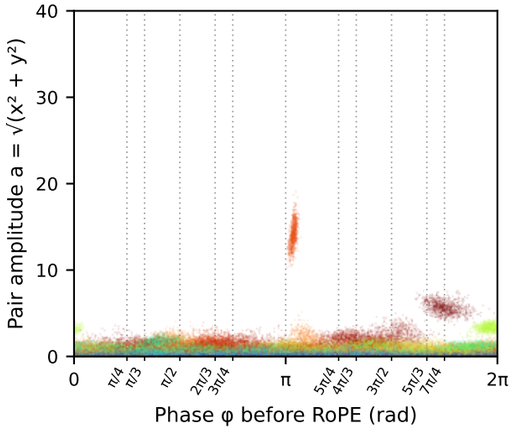
L17 H5



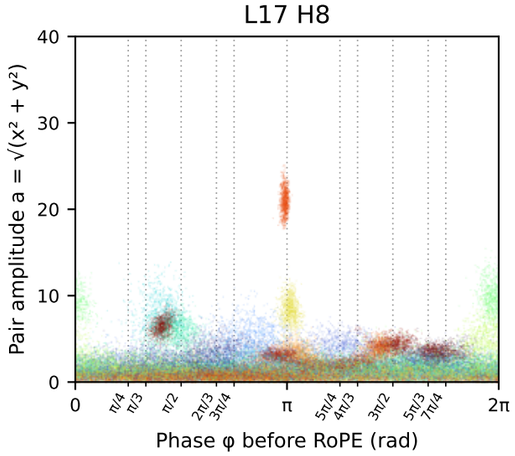
L17 H6



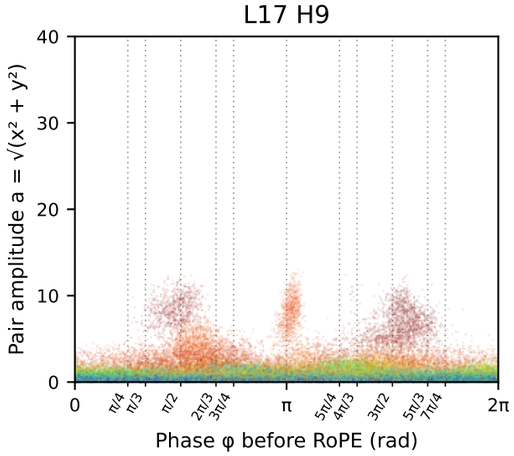
L17 H7



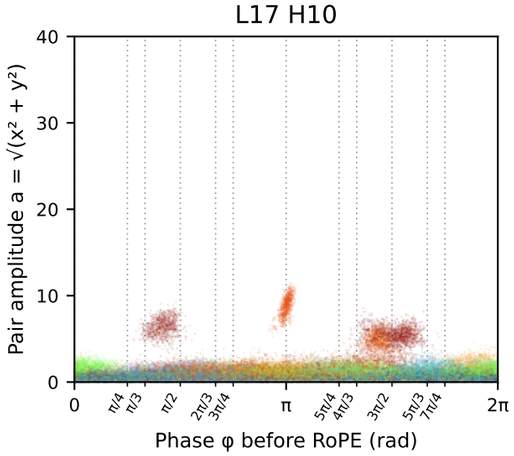
L17 H8



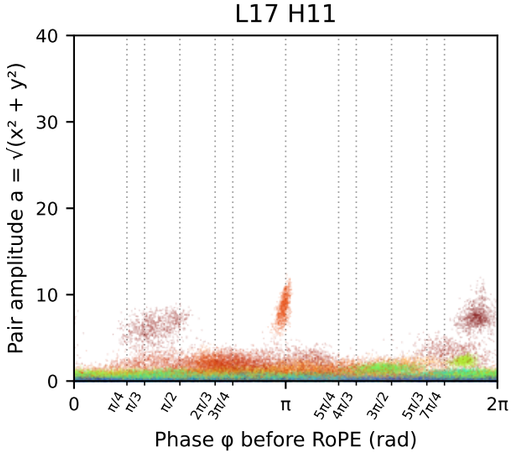
L17 H9

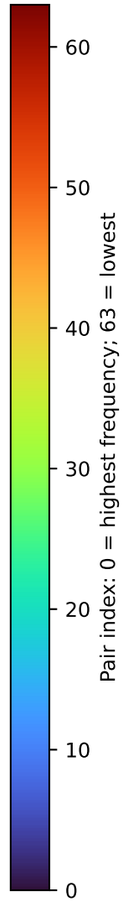


L17 H10



L17 H11

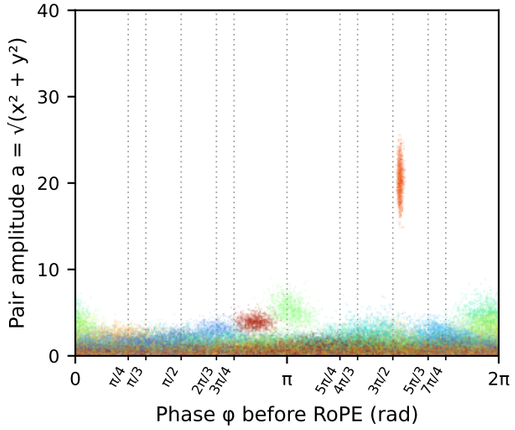




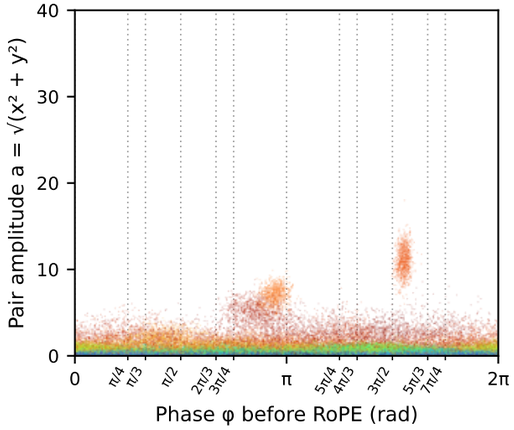
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 18 · amplitude axis 0–40

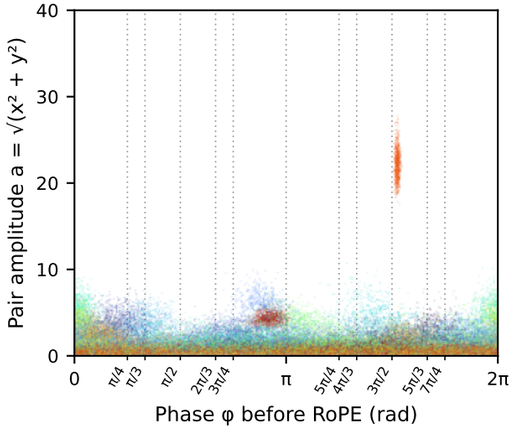
L18 H0



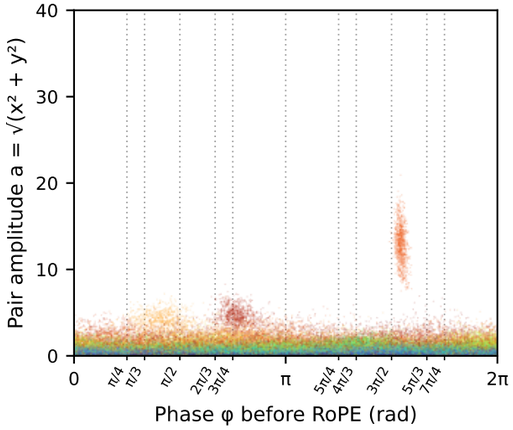
L18 H1



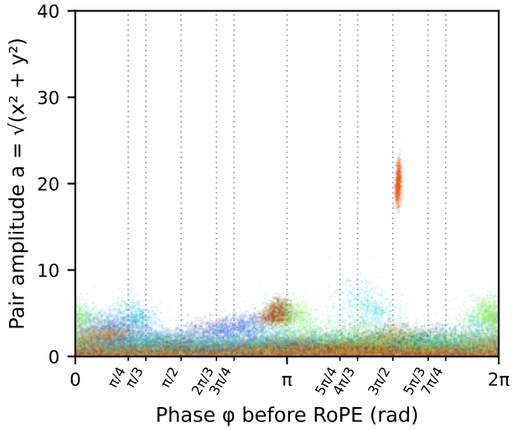
L18 H2



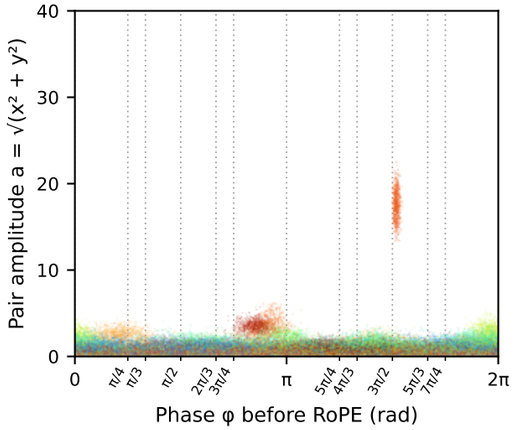
L18 H3



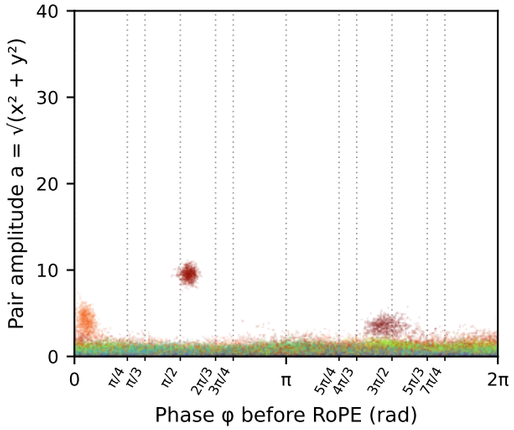
L18 H4



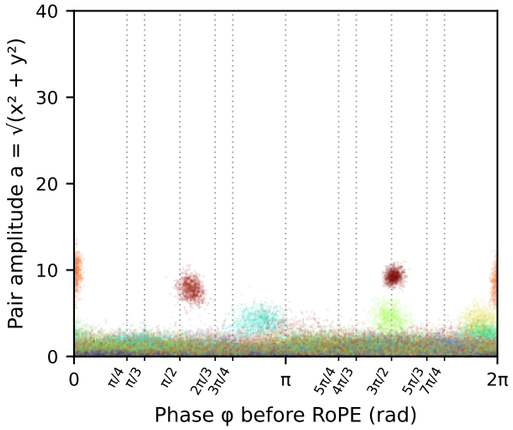
L18 H5



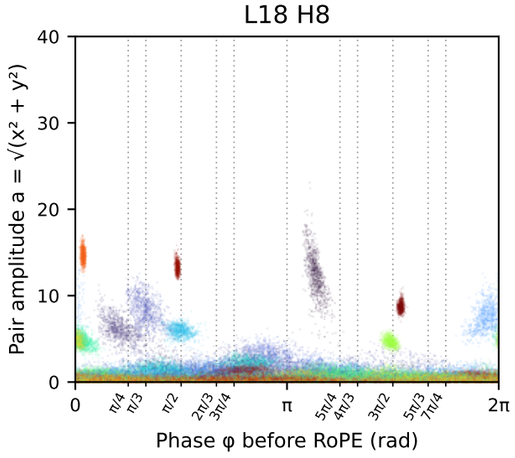
L18 H6



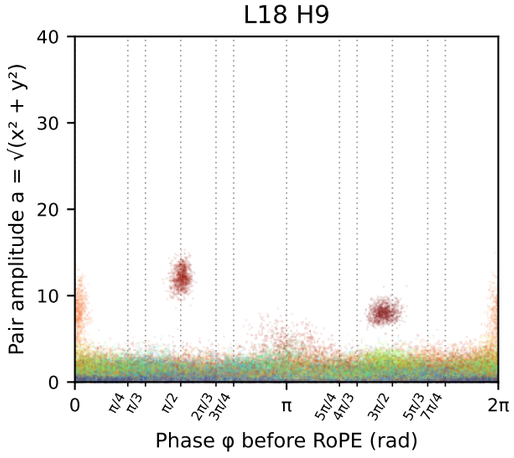
L18 H7



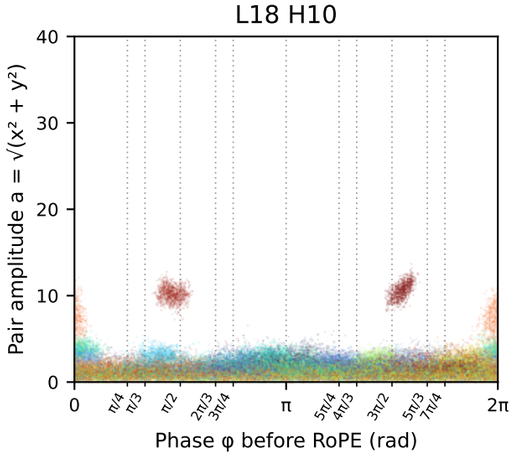
L18 H8



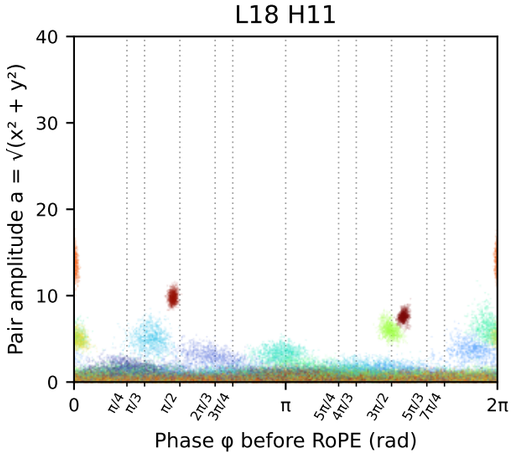
L18 H9

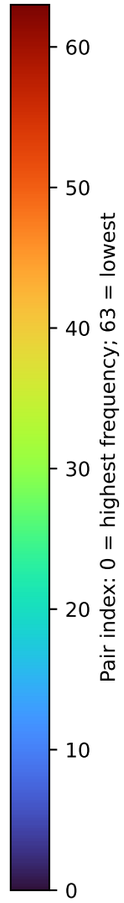


L18 H10



L18 H11

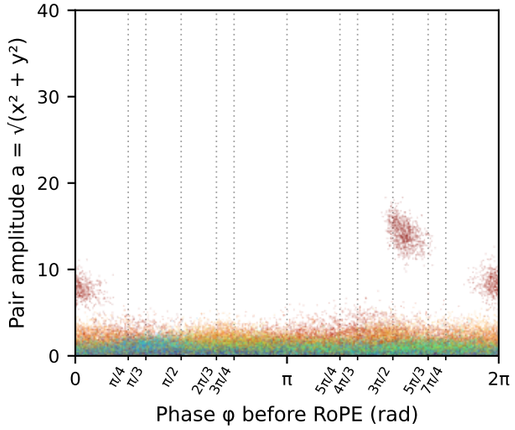




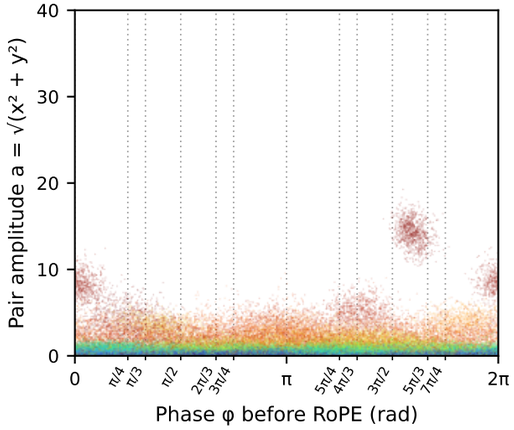
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 19 · amplitude axis 0–40

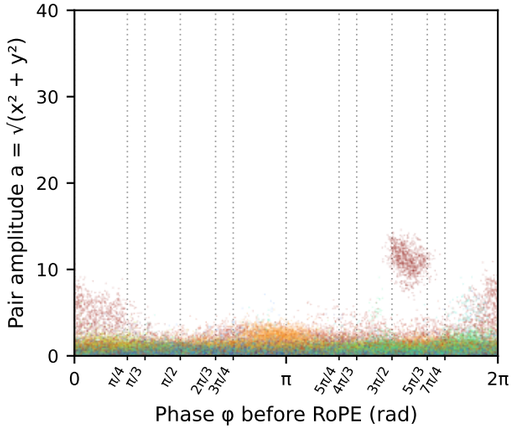
L19 H0



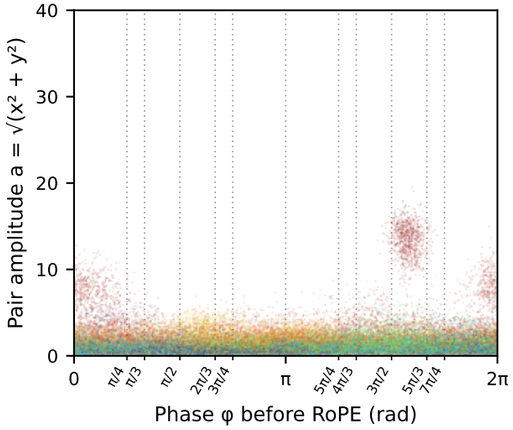
L19 H1



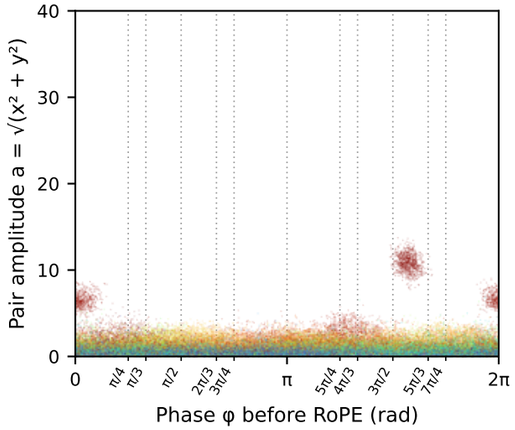
L19 H2



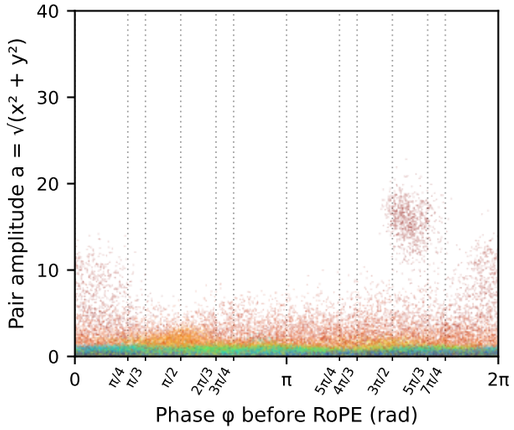
L19 H3



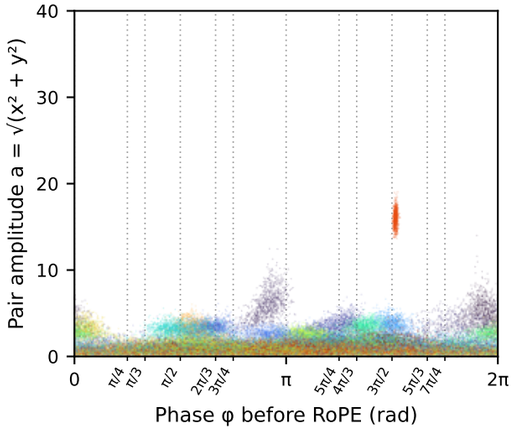
L19 H4



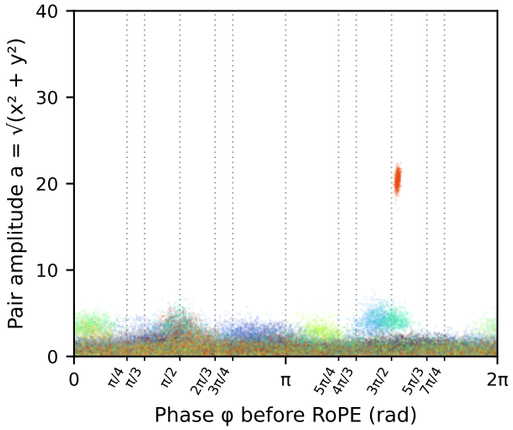
L19 H5



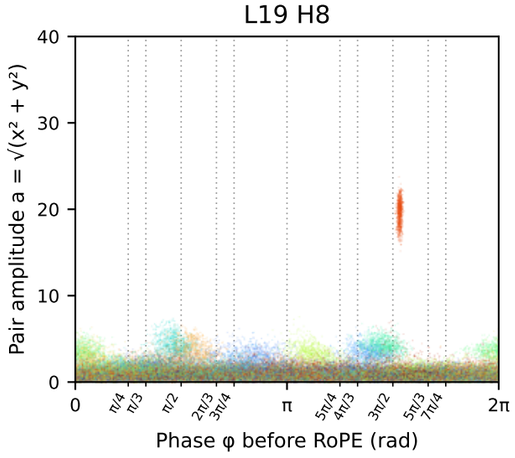
L19 H6



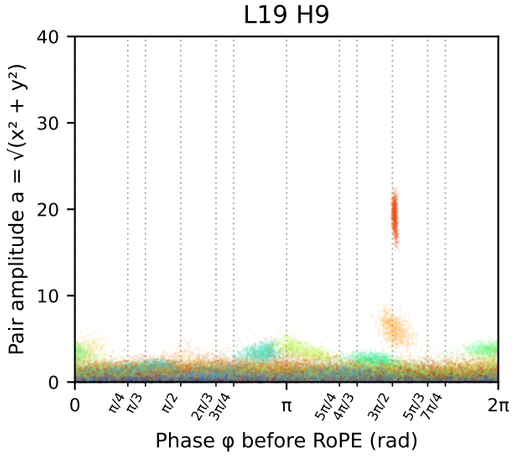
L19 H7



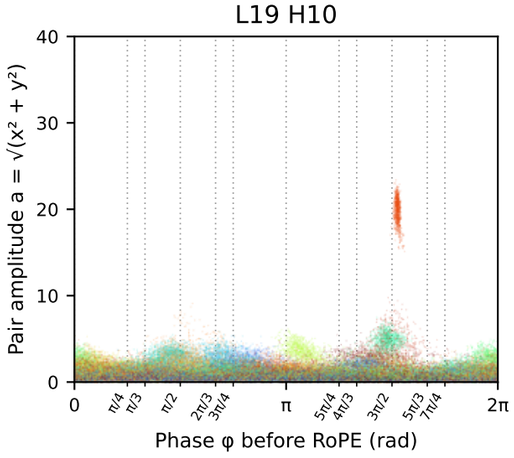
L19 H8



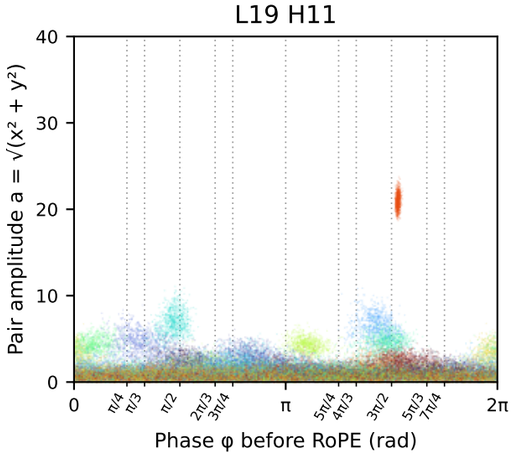
L19 H9

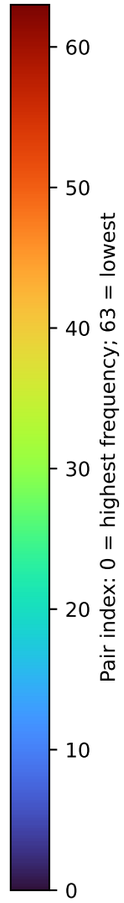


L19 H10



L19 H11

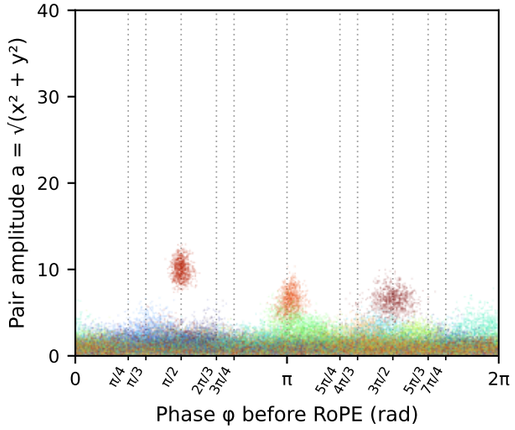




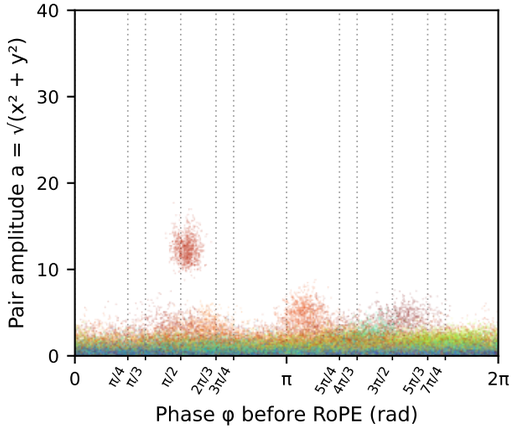
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 20 · amplitude axis 0–40

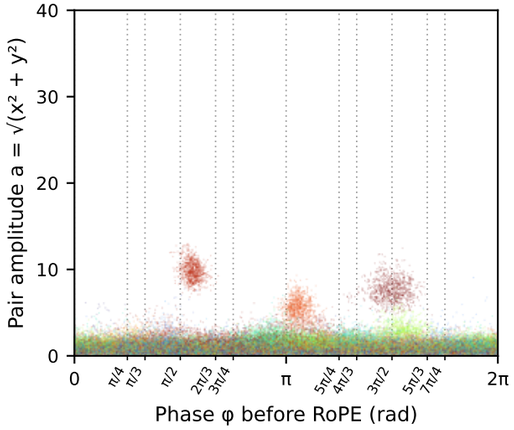
L20 H0



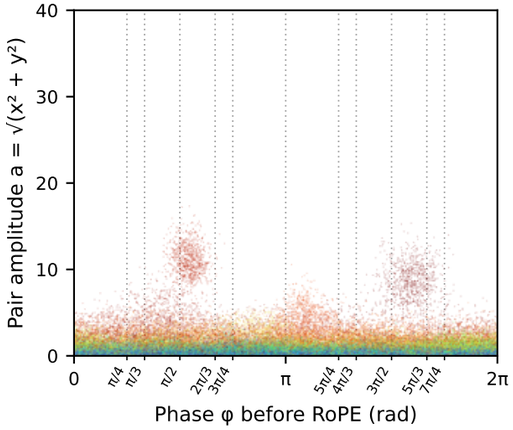
L20 H1



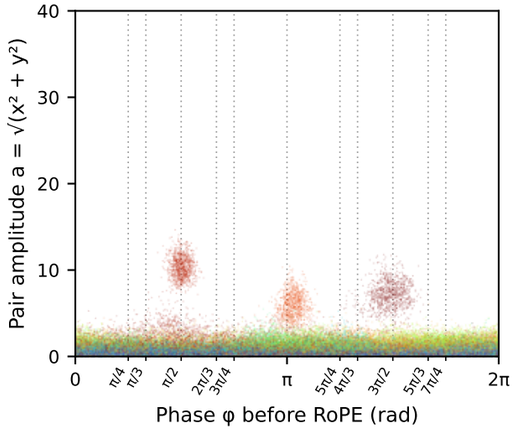
L20 H2



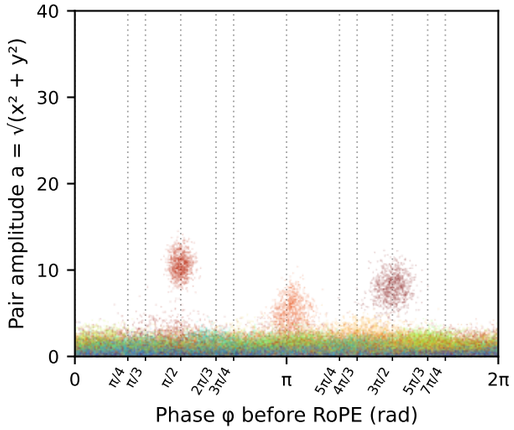
L20 H3



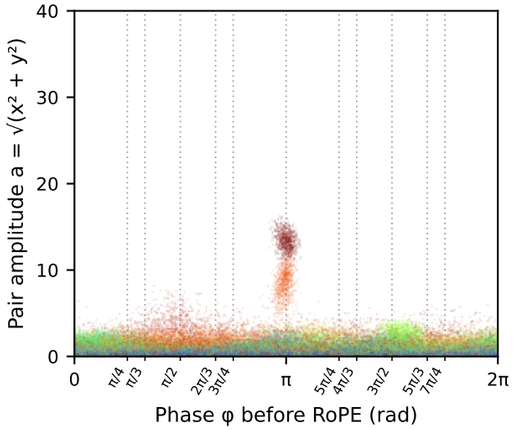
L20 H4



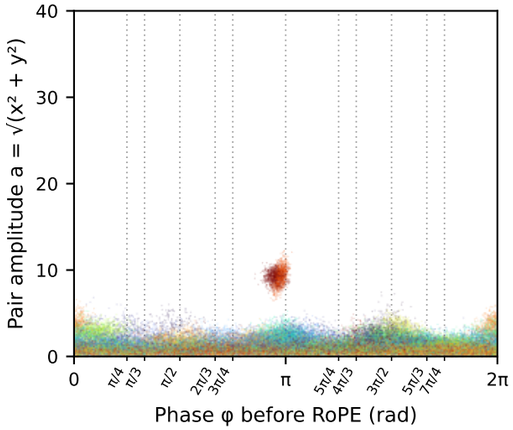
L20 H5



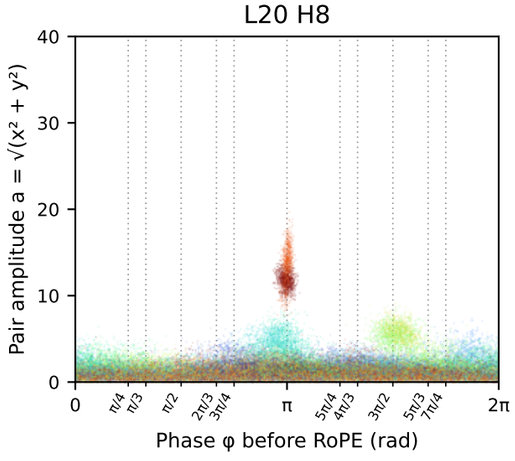
L20 H6



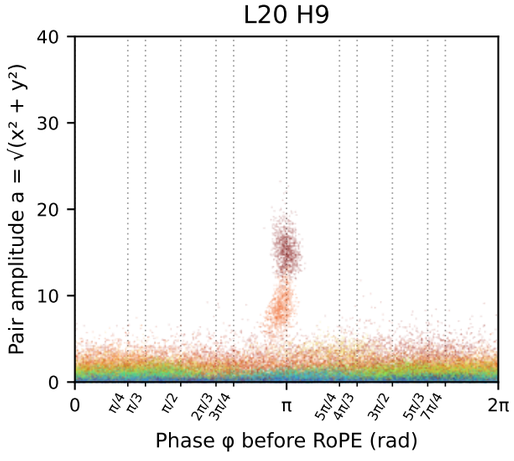
L20 H7



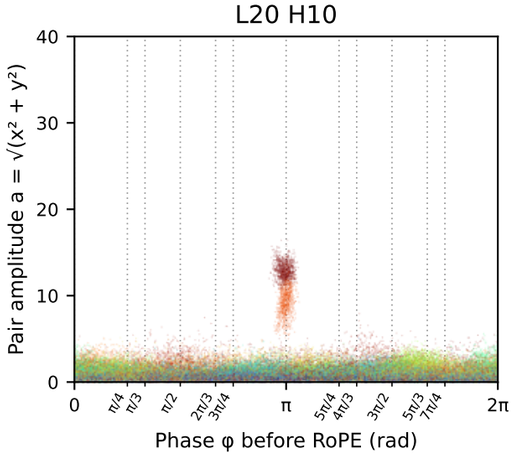
L20 H8



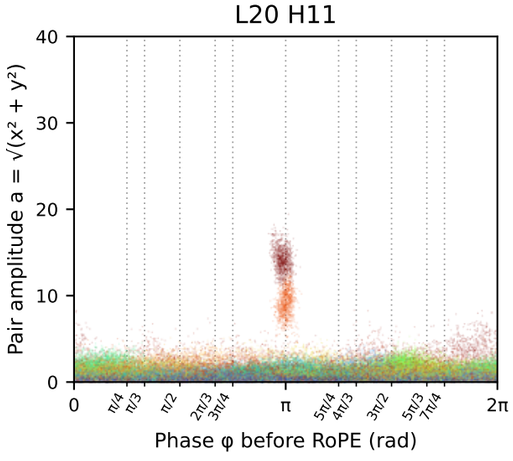
L20 H9

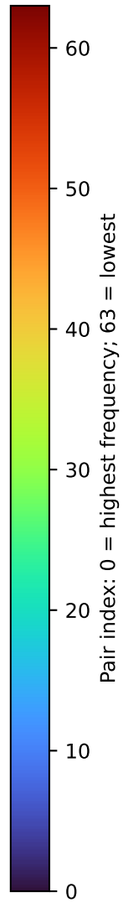


L20 H10



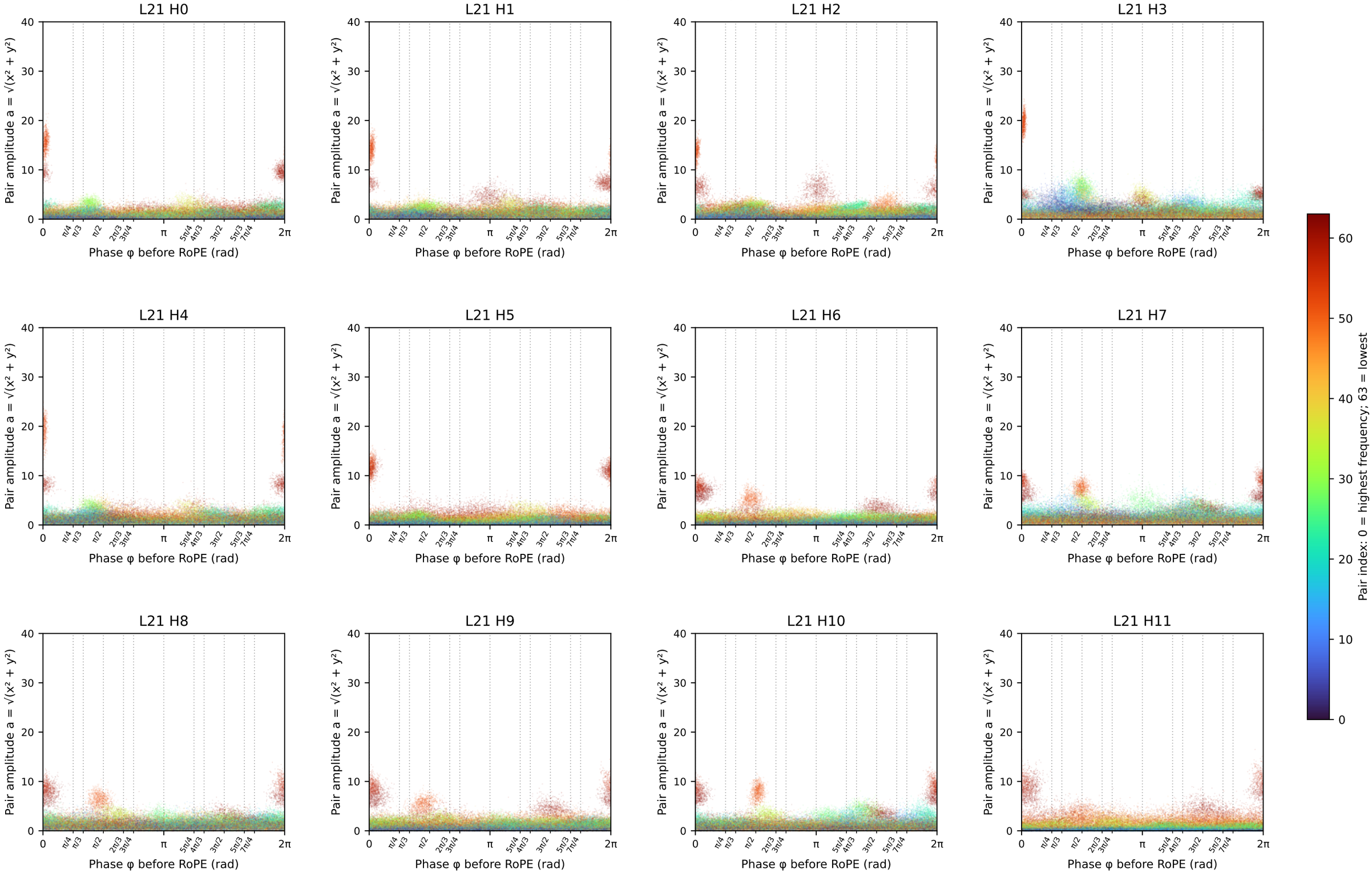
L20 H11





Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

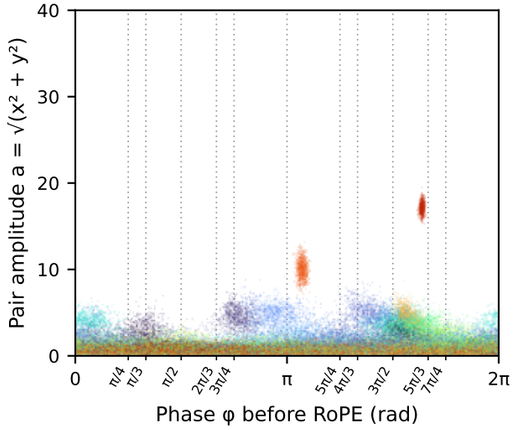
Qwen queries · layer 21 · amplitude axis 0–40



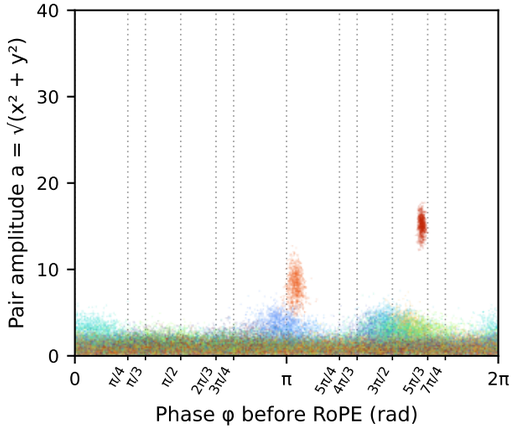
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 22 · amplitude axis 0–40

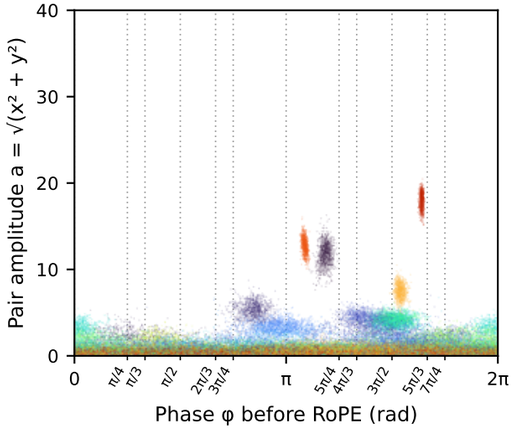
L22 H0



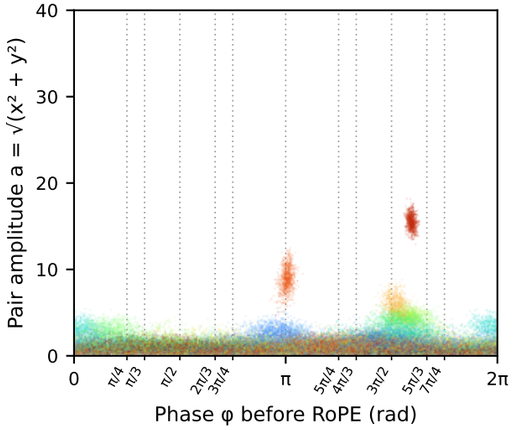
L22 H1



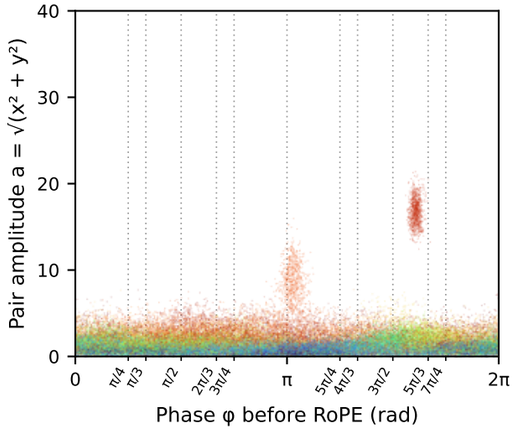
L22 H2



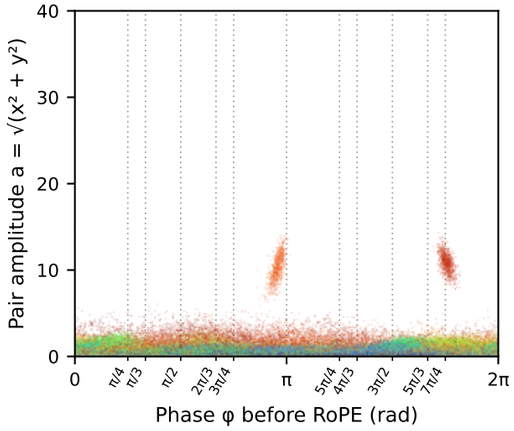
L22 H3



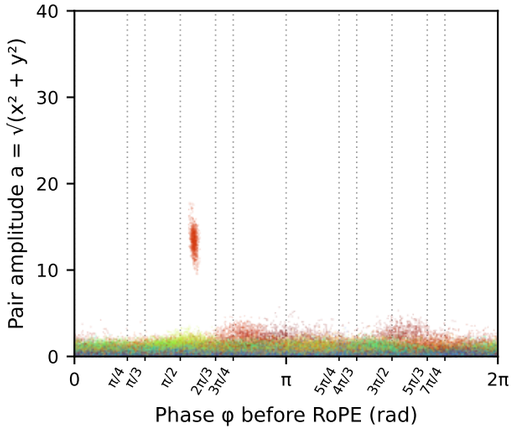
L22 H4



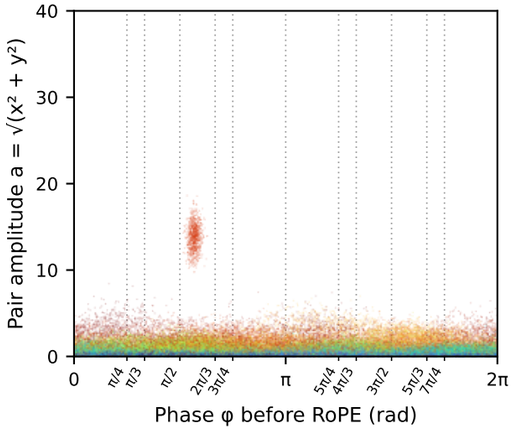
L22 H5



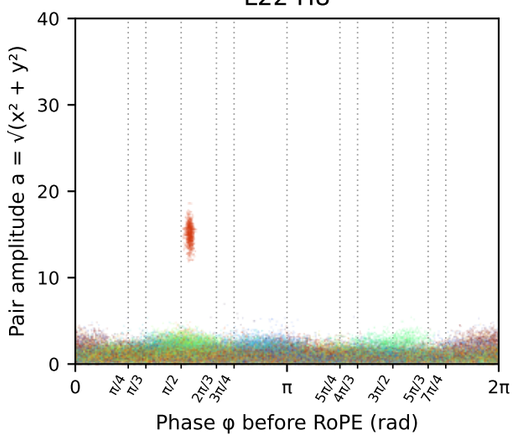
L22 H6



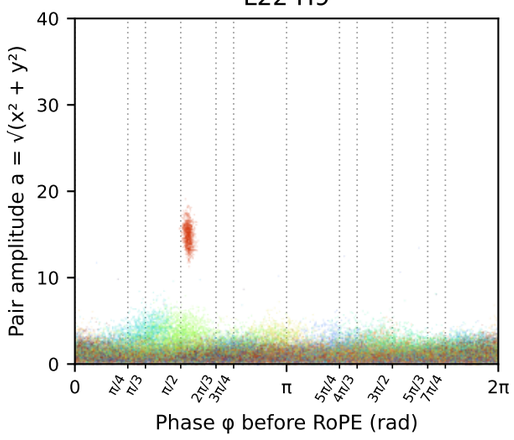
L22 H7



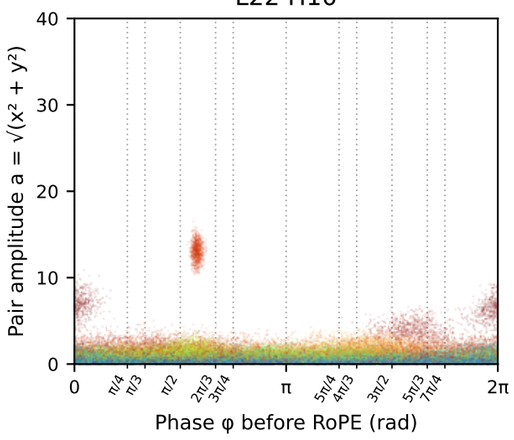
L22 H8



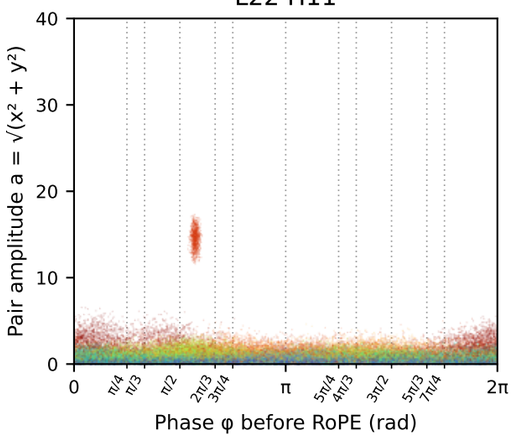
L22 H9

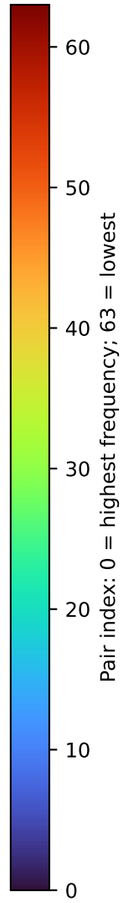


L22 H10



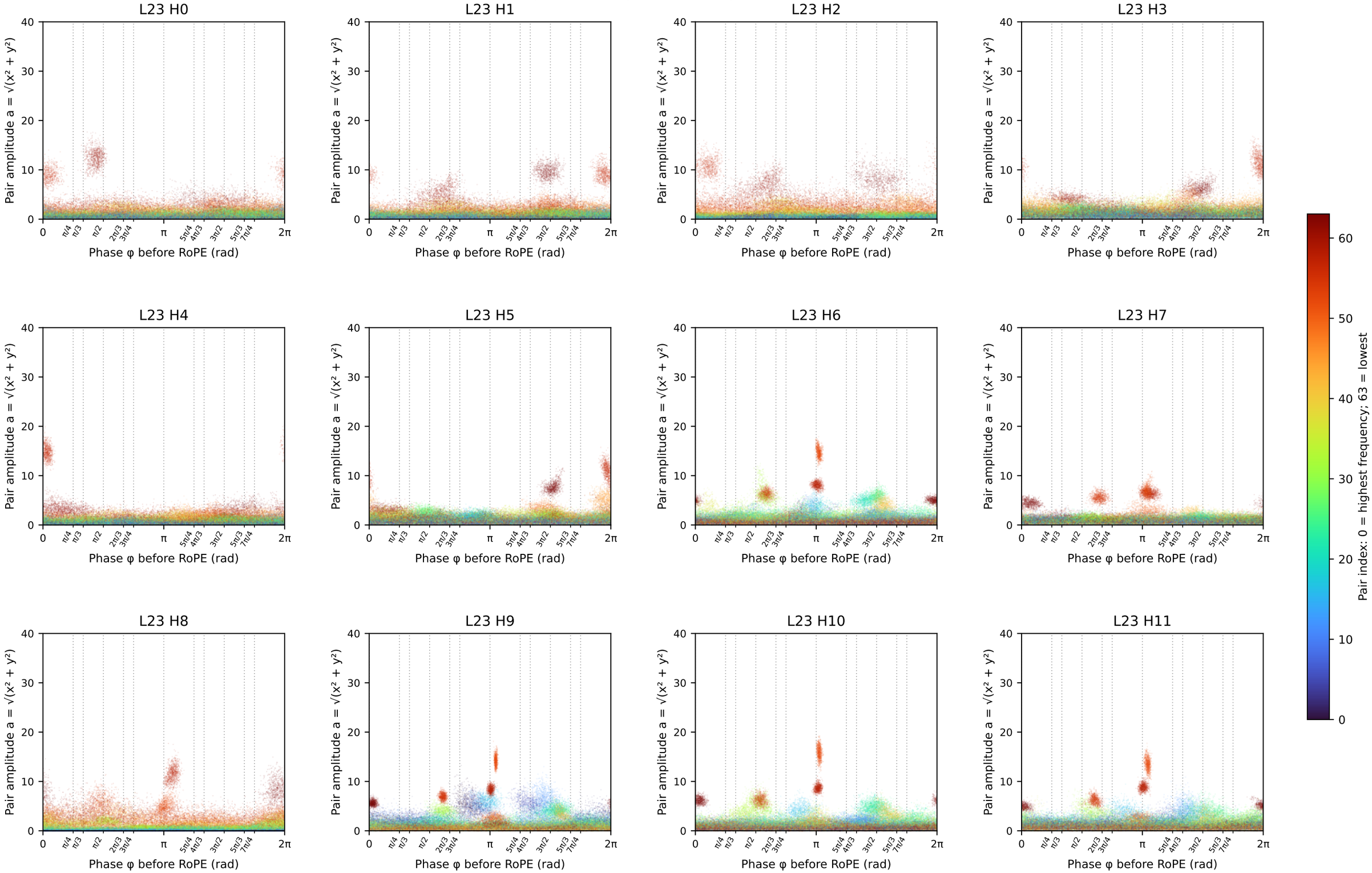
L22 H11





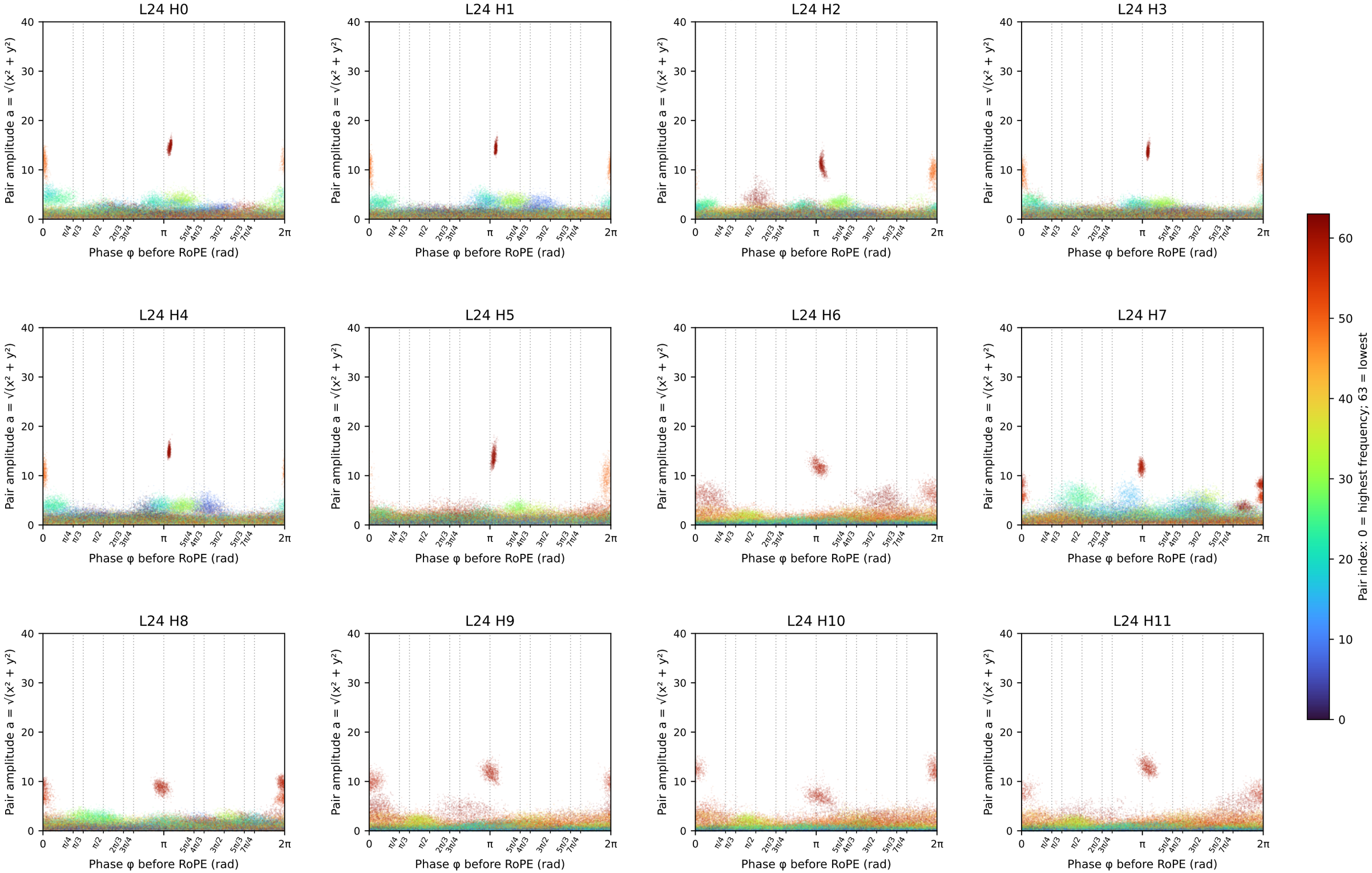
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 23 · amplitude axis 0–40



Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

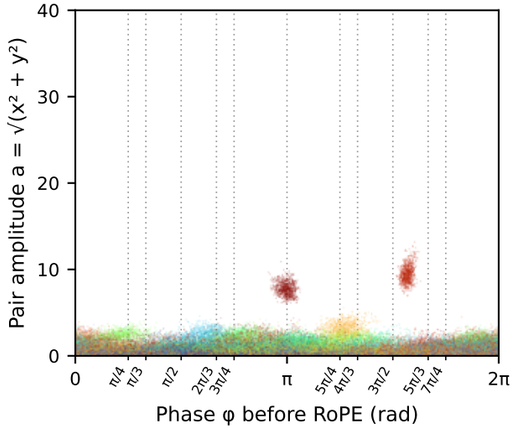
Qwen queries · layer 24 · amplitude axis 0–40



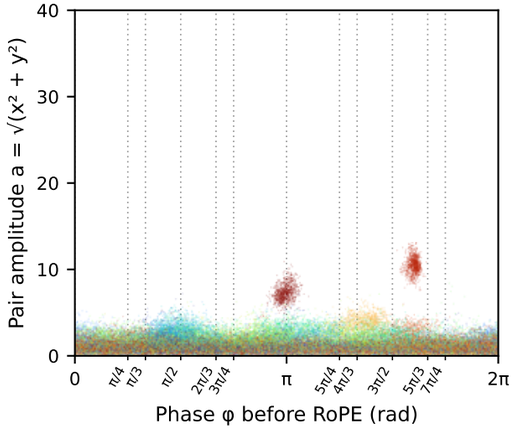
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 25 · amplitude axis 0–40

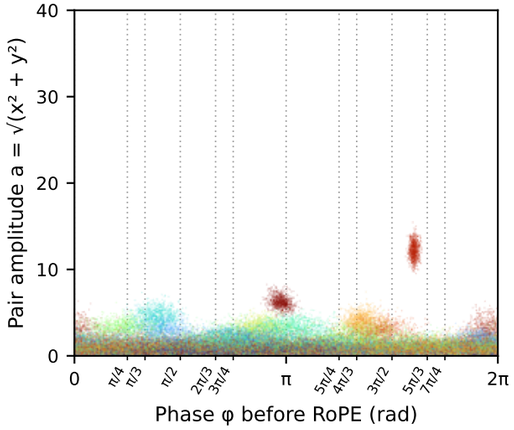
L25 H0



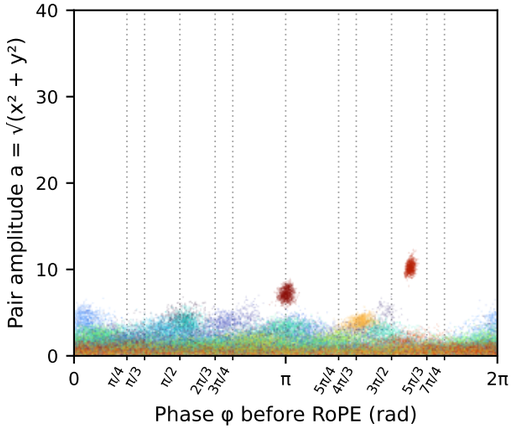
L25 H1



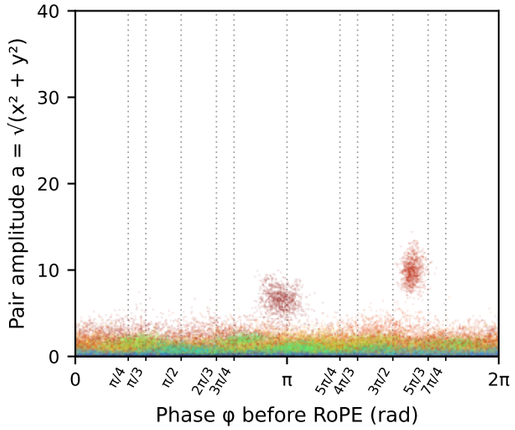
L25 H2



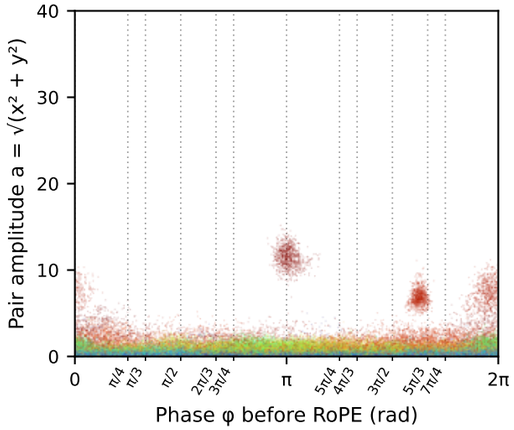
L25 H3



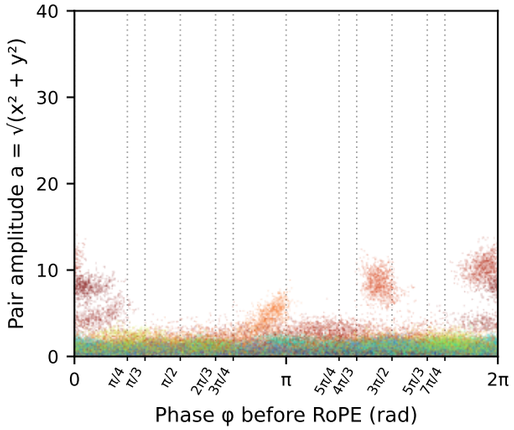
L25 H4



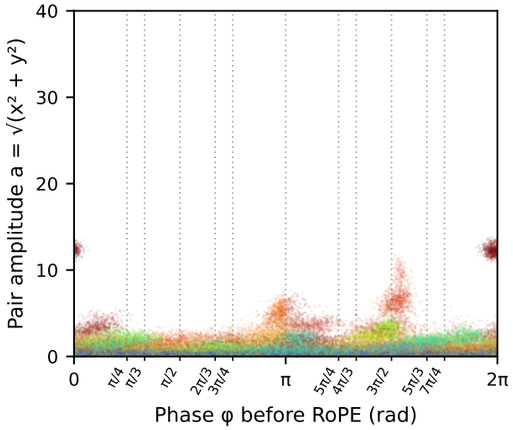
L25 H5



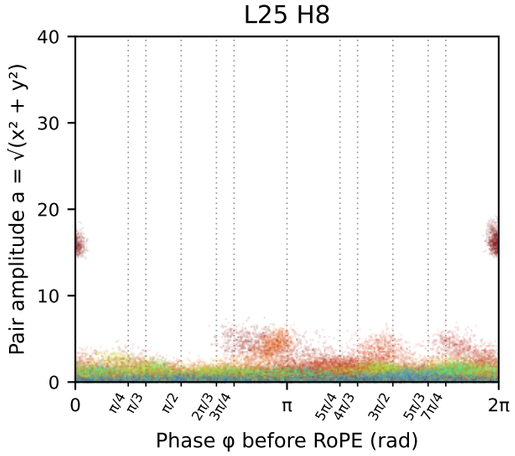
L25 H6



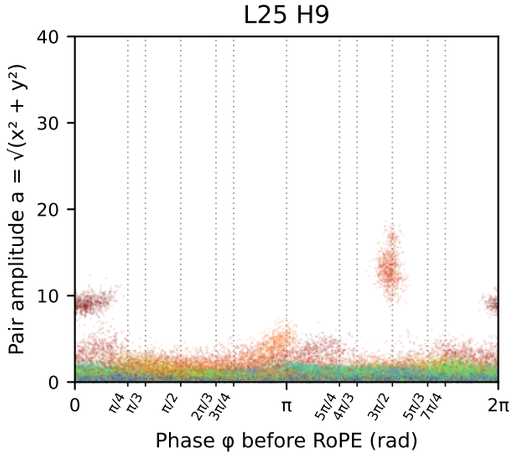
L25 H7



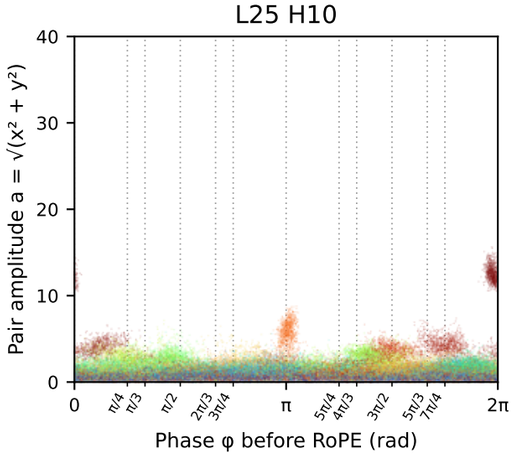
L25 H8



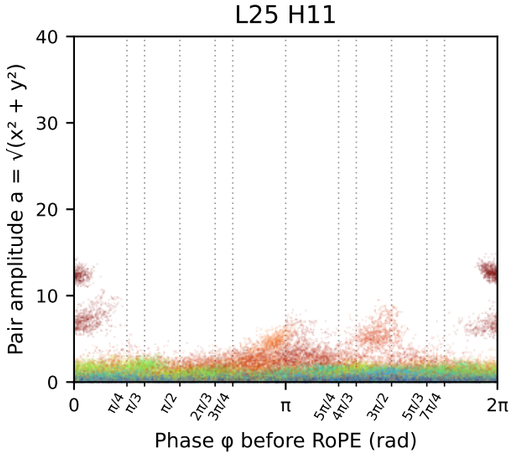
L25 H9

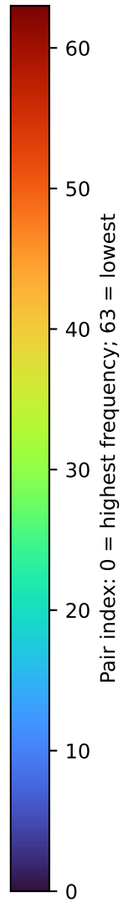


L25 H10



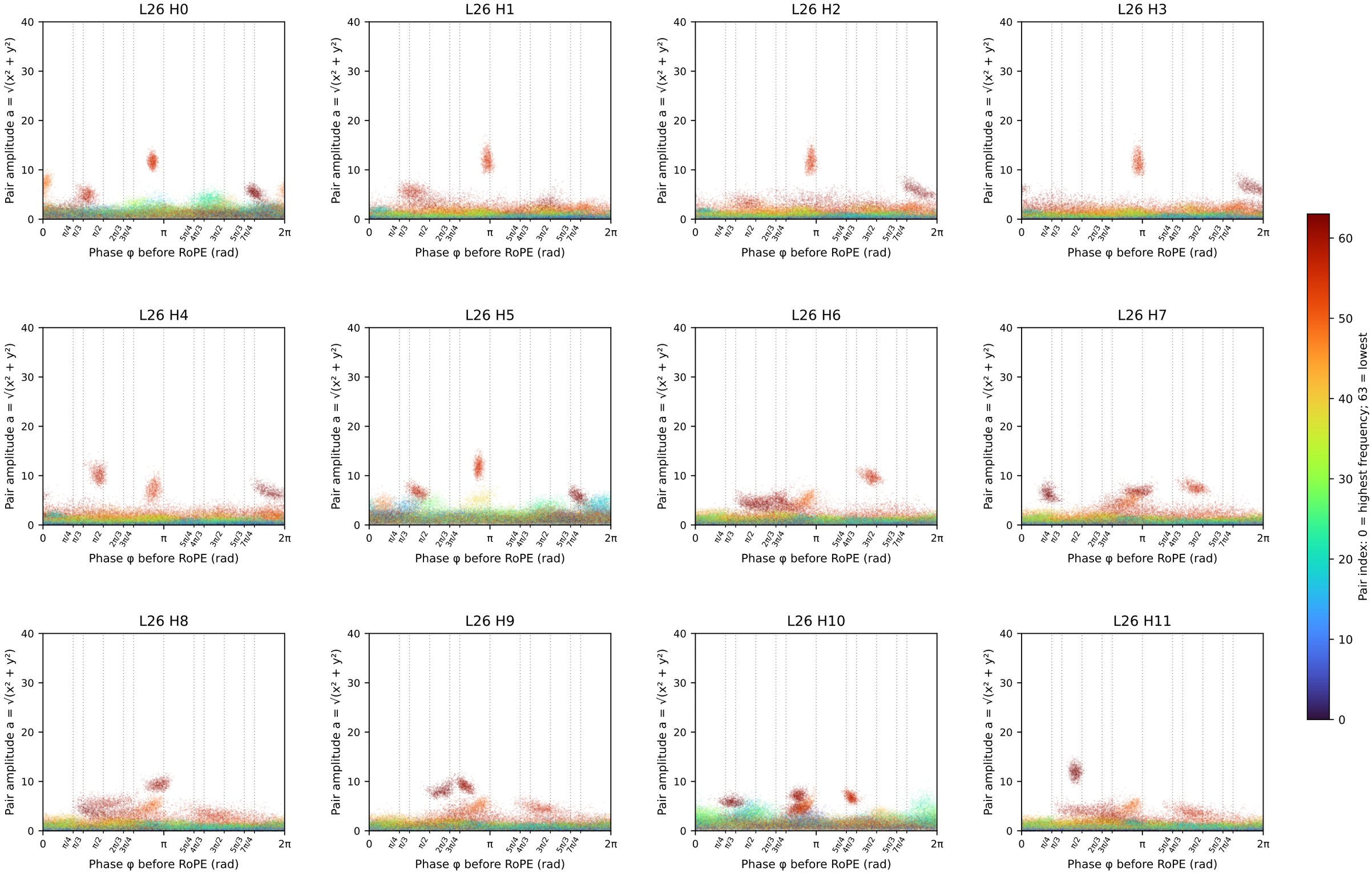
L25 H11





Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

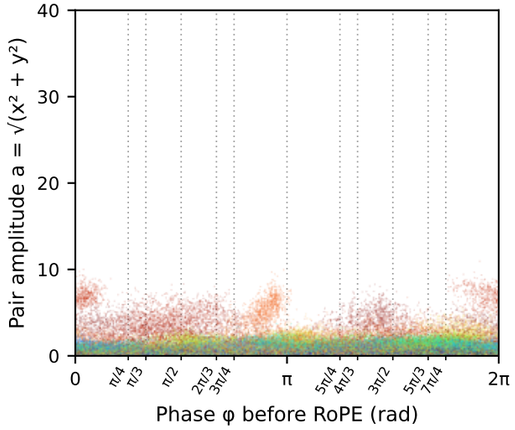
Qwen queries · layer 26 · amplitude axis 0–40



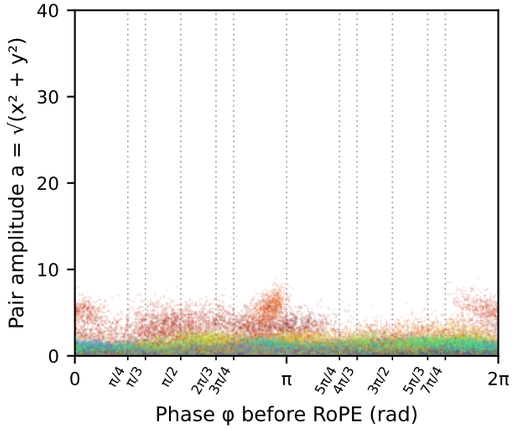
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.
Gray dotted guides mark multiples of π/4 and π/3.

Qwen queries · layer 27 · amplitude axis 0–40

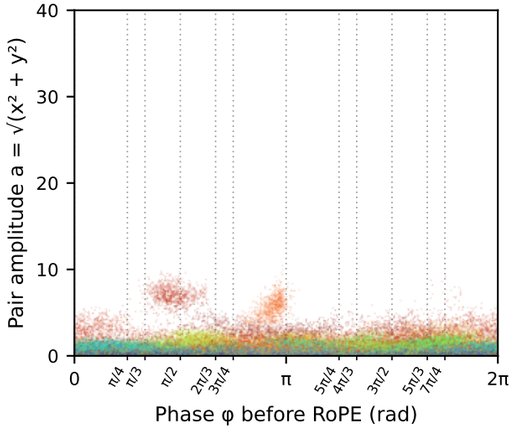
L27 H0



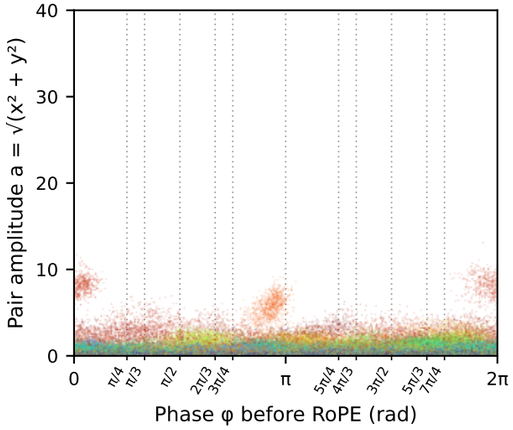
L27 H1



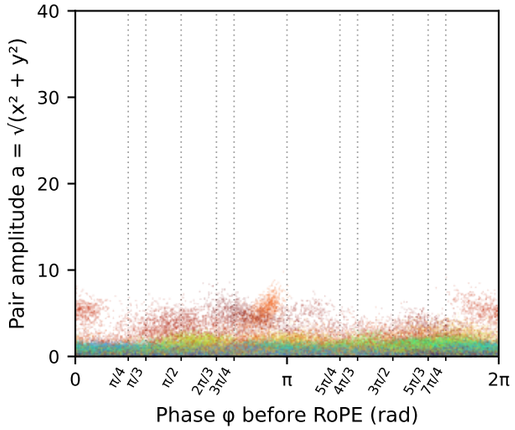
L27 H2



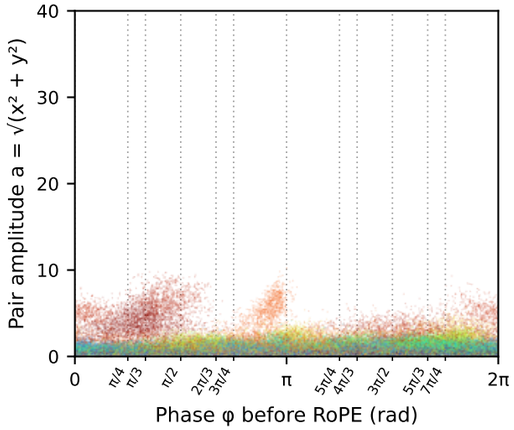
L27 H3



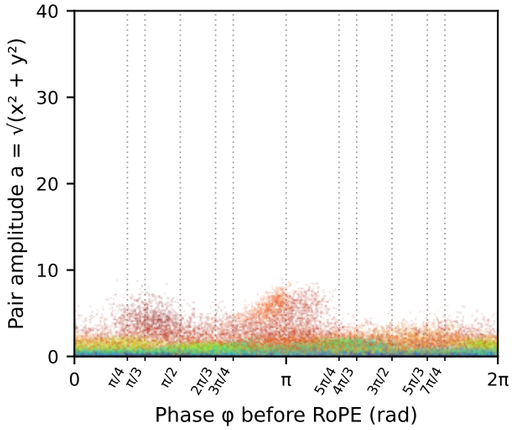
L27 H4



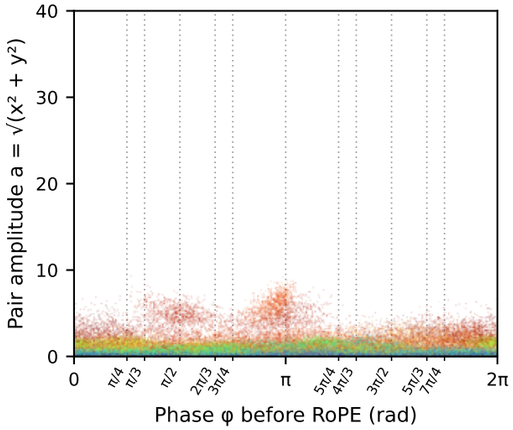
L27 H5



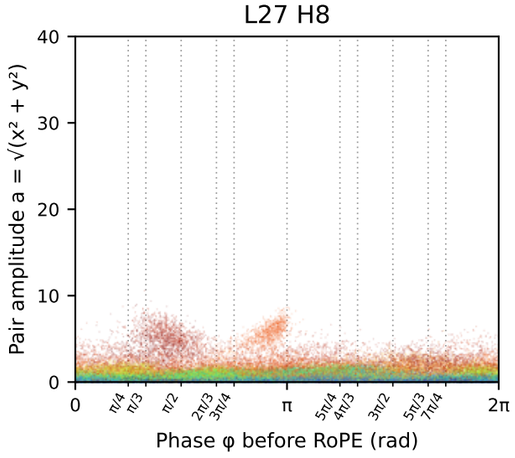
L27 H6



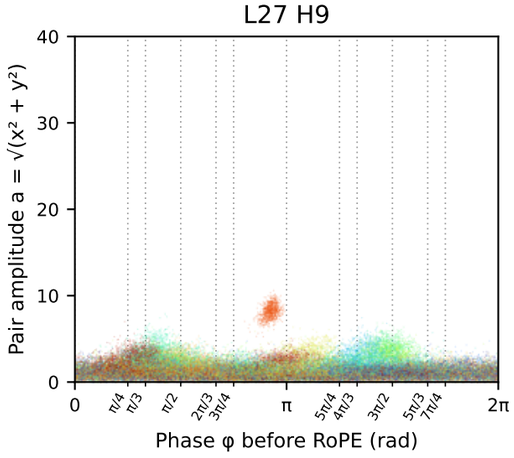
L27 H7



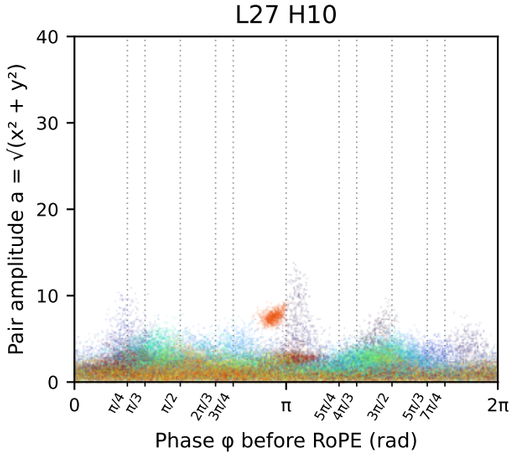
L27 H8



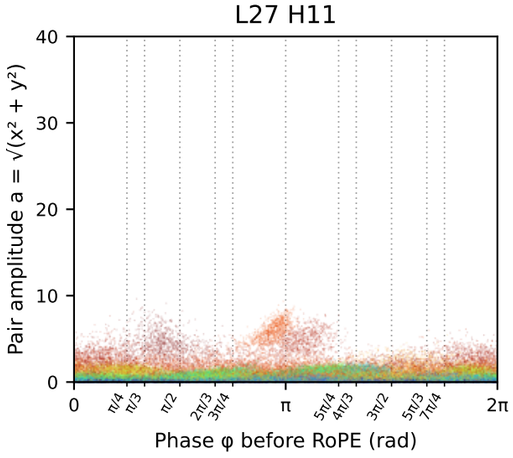
L27 H9

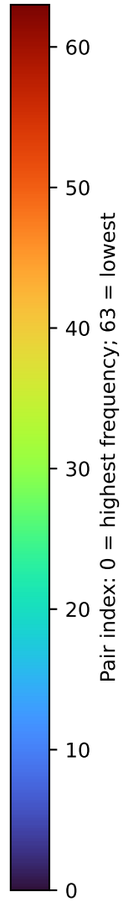


L27 H10



L27 H11





Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
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Gray dotted guides mark multiples of π/4 and π/3.